Exact additions and removals

23 September 2026

This file contains the exact source changes against `main_revised_05032026_norebutal.tex`. Blue additions and red removals are rendered in the separate full marked manuscript. The clean manuscript has 32 pages. Its SHA-256 is `b69dd56ebdbecccef7166c47376854fb815b0a83e4c790071b3b532201b2b043`.

```
--- main_revised_05032026_norebutal.tex
+++ main_revised_23092026_norebutal.tex
@@ -3,6 +3,7 @@
 \usepackage{graphicx}
 \usepackage[colorlinks,linkcolor=red]{hyperref}
 \usepackage{bigstrut,multirow,rotating}
+\usepackage{xcolor,array,tabularx,xurl,placeins}
 \usepackage{booktabs}
 \usepackage{enumerate}
 \usepackage{amsmath}
@@ -14,6 +15,7 @@
 \usepackage{authblk} % <-- Add this for multiple authors/affiliations
 \usepackage[mathlines]{lineno}
 \linenumbers
+\setlength{\emergencystretch}{2em}
 % --- Theorem styles ------------------------------------------------
 \theoremstyle{plain}      % default: bold header + italic body
 \newtheorem{theorem}{Theorem}[section]     % theorem 3.2, 3.3, ... within section 3
@@ -44,7 +46,7 @@
 \affil[5]{The Alan Turing Institute, London, UK}


-\date{}
+\date{23 September 2026}

 \begin{document}

@@ -52,11 +54,9 @@


 \begin{abstract}
-We present a data-driven, kernel-based framework for reconstructing the
↪  quasi-potential of stochastic differential equations (SDEs) in the weak-noise
↪  regime.
-The method approximates a family of Koopman eigenfunctions in a reproducing-kernel
↪  Hilbert space and combines them into a closed-form surrogate whose gradient
↪  automatically satisfies a quadratic Hamilton-Jacobi ansatz.
-A composite loss simultaneously (i) minimises the Hamilton-Jacobi residual on a
↪  collocation grid, (ii) matches the Hessian of the surrogate to the solution of the
↪  Riccati equation at the equilibrium, and (iii) promotes sparsity through an
↪  \(\ell_{1}\) penalty on the kernel-mixture weights.
```

```
-
-Numerical experiments show that the analytic surrogate is inexpensive to evaluate,
 ↪  paving the way for efficient rare-event analysis and optimal-control applications
 ↪  in higher-dimensional, non-gradient stochastic systems.
+We present a kernel framework for computing quasi-potentials of stochastic
 ↪  differential equations in the weak-noise regime. Direct scalar RKHS approximations
 ↪  and structured Koopman-coordinate surrogates combine Hamilton--Jacobi residual
 ↪  fitting with diffusion-dependent Riccati curvature at a stable equilibrium.
 ↪  Kernel-amplitude regularization and finite-dictionary selection provide
 ↪  complementary ways to control representation complexity. A conditional convergence
 ↪  theorem states the compactness, residual-control, and boundary comparison
 ↪  assumptions needed to identify the viscosity limit.
+
+Numerical experiments recover two quartic reference potentials with polynomial kernels
 ↪  and reproduce a three-dimensional multiplicative-noise quadratic with a stable
 ↪  anchored kernel expansion. The three-dimensional fit achieves an RMSE of
 ↪  $1.35\times10^{-9}$ on an $8000$-point grid at $N=200$. Independent tests, multiple
 ↪  training seeds for the two-dimensional systems, and a structured eight-dimensional
 ↪  experiment assess accuracy and computational cost. A comparison with restricted
 ↪  quadratic Koopman representations identifies approximation bias associated with
 ↪  transformed diffusion. All numerical results are accompanied by code, fitted
 ↪  coefficients, and native vector figures.
 \end{abstract}

 \section{Introduction}
@@ -67,9 +67,12 @@

 Recent advances in computational power and artificial intelligence have spurred the
  ↪  application of machine learning to stochastic dynamics. Techniques such as sparse
  ↪  regression \cite{Boninsegna2018, YangLi2020a}, Koopman operators \cite{EDMD,
  ↪  SKO3}, and neural networks \cite{XiaoliChen} have been leveraged for data-driven
  ↪  modeling of stochastic systems. In addition, some machine learning methods are
  ↪  also proposed to solve the problem of noise-induced transitions such as computing
  ↪  the most probable paths \cite{li2021machine, wei2022optimal} and the mean first
  ↪  exit times \cite{li2024deep}. Notably, Lin et al. \cite{lin2022data} proposed
  ↪  neural network-based frameworks to compute quasi-potentials for stochastic
  ↪  dynamical systems with additive Gaussian noise, while Li et al.
  ↪  \cite{li2022machine} developed physics-informed neural networks (PINNs) to compute
  ↪  quasi-potentials by solving Hamilton-Jacobi equations.

-Among machine learning paradigms, kernel methods stand out due to their strong
 ↪  nonlinear representation capabilities and computational efficiency. Notably,
 ↪  reproducing kernel Hilbert spaces (RKHS) \cite{CuckerandSmale} have established
 ↪  solid mathematical foundations for the analysis of  dynamical systems \cite{BH4,
 ↪  yk1, bhcm11,bhcm1, lyap_bh, bh_sparse_kfs, BHPhysicaD, hamzi2019kernel,
 ↪  bh2020b,SKO3,ALEXANDER2020132520,bh12,bh17,hb17,mmd_kernels_bh,
 ↪  hamzipaillet,jalalian_deds_2023,lee2024note, hamzi2025kernel_lions,
 ↪  bittracher2021dimensionality} and surrogate modeling (cf.\ \cite{santinhaasdonk19}
 ↪  for a survey) as well as analyzing neural networks \cite{smirnov2022mean}.
 ↪  \textcolor{red}{The idea of combining multiple kernels by learning a weighted
 ↪  mixture is known as \emph{multiple kernel learning} (MKL)
 ↪  \cite{lanckriet2004learning, bach2004multiple} and has been studied extensively in
 ↪  the machine learning literature. Our approach is related to MKL in that we learn a
 ↪  sparse convex combination of base kernels, but it differs in that the objective
 ↪  couples kernel selection with PDE residual minimization and curvature matching,
 ↪  rather than regression or classification loss.}
-
```

```
-In this work, we present a kernel-based algorithm for quasi-potential computation. The
 ↪  structure of this paper is as follows: \textcolor{red}{Section~\ref{qpsec}
 ↪  introduces the definition and properties of the quasi-potential.
 ↪  Section~\ref{methsec} details the proposed kernel-driven methodology. Numerical
 ↪  examples in Section~\ref{examples} validate the algorithm's accuracy and
 ↪  efficiency, followed by concluding remarks.}
+Among machine learning paradigms, kernel methods stand out due to their strong
 ↪  nonlinear representation capabilities and computational efficiency. Notably,
 ↪  reproducing kernel Hilbert spaces (RKHS) \cite{CuckerandSmale} have established
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 ↪  yk1, bhcm11,bhcm1, lyap_bh, bh_sparse_kfs, BHPhysicaD, hamzi2019kernel,
 ↪  bh2020b,SKO3,ALEXANDER2020132520,bh12,bh17,hb17,mmd_kernels_bh,
 ↪  hamzipaillet,jalalian_deds_2023,lee2024note, hamzi2025kernel_lions,
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 ↪  for a survey) as well as analyzing neural networks \cite{smirnov2022mean}. The idea
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 ↪  \emph{multiple kernel learning} (MKL) \cite{lanckriet2004learning,
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 ↪  literature. Our approach is related to MKL in that we learn a sparse convex
 ↪  combination of base kernels, but it differs in that the objective couples kernel
 ↪  selection with PDE residual minimization and curvature matching, rather than
 ↪  regression or classification loss.
+
+\paragraph{Position relative to existing approaches.}\label{sec:positioning}
+Kernel collocation and Gaussian-process methods provide smooth representations,
 ↪  derivative evaluation, and nonlinear PDE residual fitting \cite{Chen2021}.
 ↪  Physics-informed quasi-potential methods impose the Hamilton--Jacobi equation
 ↪  \cite{li2022machine,lin2022data}, while kernel Koopman methods approximate
 ↪  dynamical eigenfunctions and spectra \cite{Williams2015}. The present contribution
 ↪  couples a diffusion-dependent Riccati curvature target with direct scalar and
 ↪  Koopman-coordinate kernel surrogates, and specifies their distinct approximation
 ↪  properties. The Riccati term supplies local nontrivial geometry; global viscosity
 ↪  selection requires further conditions. Koopman coordinates give a structured
 ↪  nonnegative parametrization and linear inner transport fits, but their benefit over
 ↪  direct scalar RKHS fitting is problem dependent. Section~\ref{sec:koopman-role}
 ↪  explains this distinction, and Appendix~\ref{app:additional} provides a controlled
 ↪  comparison. Section~\ref{examples} studies two- and three-dimensional systems.
 ↪  Neither the meshfree representation nor those examples establishes general
 ↪  high-dimensional scalability.
+
+In this work, we present a kernel-based algorithm for quasi-potential computation. The
 ↪  structure of this paper is as follows: Section~\ref{qpsec} introduces the
 ↪  definition and properties of the quasi-potential. Section~\ref{methsec} details the
 ↪  proposed kernel-driven methodology. Numerical examples in Section~\ref{examples}
 ↪  examine reconstruction accuracy and computational limitations, followed by
 ↪  concluding remarks.



@@ -98,37 +101,33 @@


 \setcounter{equation}{0}
-\section{Quasi-Potential and the Wentzel-Kramers-Brillouin Approximation in Stochastic
 ↪  Dynamical Systems} \label{qpsec}
+\section{Quasi-Potential and the Wentzel--Kramers--Brillouin Approximation in
 ↪  Stochastic Dynamical Systems} \label{qpsec}
```

```
 In this section, we examine dynamical systems governed by state variables \( x_t \in
 ↪  \mathbb{R}^n \), evolving under the influence of Gaussian white noise with low
 ↪  intensity. The dynamics are described by the stochastic differential equation:
 \begin{equation}
 \label{sde}
 dx_t = b(x_t)\,dt + \sqrt{\varepsilon}\,\sigma(x_t)\,dB_t,
 \end{equation}
-\iffalse
-where \( B_t \in \mathbb{R}^n \) denotes an \( n \)-dimensional standard Brownian
↪  motion, and \( \varepsilon \) is a small positive parameter quantifying the noise
↪  strength. \textcolor{red}{The function \( b(x) \in \mathbb{R}^n \) represents the
↪  deterministic drift term which has continuously differentiable derivatives,} while
↪  the diffusion coefficient is given by the matrix \( a(x) = \sigma(x)\sigma(x)^T \).
-
-% REPLACE existing text after Equation (2) with:
-\fi
+
 where $B_t \in \mathbb{R}^n$ denotes an $n$-dimensional standard Brownian motion, and
 ↪  $\varepsilon$ is a small positive parameter quantifying the noise strength. The
 ↪  function $b(x) \in \mathbb{R}^n$ represents the deterministic drift term, which we
 ↪  assume has continuously differentiable derivatives. The diffusion coefficient is
 ↪  given by the matrix $a(x) = \sigma(x)\sigma(x)^\top$, where we assume $a(x)$ is
 ↪  uniformly positive definite on $\Omega$ (ensuring non-degeneracy of the noise).

-We suppose $\bar{x}$ is an \textbf{asymptotically stable and hyperbolic} fixed point
↪  of the associated deterministic system, that is, $b(\bar{x}) = 0$ and all
↪  eigenvalues of its linearization matrix $J = \nabla b(\bar{x})$ have
↪  \textbf{strictly negative real parts}, i.e., $\text{Re}(\lambda_i) < 0$ for all $i
↪  = 1,\ldots,n$. This ensures asymptotic stability in the sense that trajectories
↪  starting near $\bar{x}$ converge to $\bar{x}$ exponentially in time. Under
↪  stochastic perturbations, the process $x_t$ explores the state space and converges
↪  to a stationary probability distribution $p_s(x)$. In the regime of weak noise
↪  ($\varepsilon \ll 1$), this stationary distribution admits a
↪  Wentzel--Kramers--Brillouin (WKB)-type asymptotic approximation (see
↪  \cite{chen2017singularities,li2022machine,smelyanskiy1997topological}):
+We suppose $\bar{x}$ is an \textbf{asymptotically stable and hyperbolic} fixed point
↪  of the associated deterministic system, that is, $b(\bar{x}) = 0$ and all
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↪  \textbf{strictly negative real parts}, i.e., $\text{Re}(\lambda_i) < 0$ for all $i
↪  = 1,\ldots,n$. This ensures asymptotic stability in the sense that trajectories
↪  starting near $\bar{x}$ converge to $\bar{x}$ exponentially in time. Under
↪  additional global confinement and recurrence assumptions, the perturbed process
↪  admits a stationary density $p_s(x)$; local stability alone does not imply this
↪  global property. In the regime of weak noise ($\varepsilon \ll 1$), this stationary
↪  distribution admits a Wentzel--Kramers--Brillouin (WKB)-type asymptotic
↪  approximation (see
↪  \cite{chen2017singularities,li2022machine,smelyanskiy1997topological}):
 \begin{equation}
 \label{WKB}
 p_s(x) \sim C(x)\, \exp\left\{-\varepsilon^{-1} V(x)\right\},
 \end{equation}
```

```
 where \( C(x) \) is a slowly varying prefactor and \( V(x) \) is the quasi-potential
 ↪  function, characterizing the exponential decay rate of the probability density.
 ↪  This form is termed a \emph{WKB-type approximation} named after the WKB method in
 ↪  quantum mechanics, where solutions in the semiclassical limit \( \hbar \to 0 \)
 ↪  take the form \( \psi(x) \sim A(x)\exp\{i S(x)/\hbar\} \). In both contexts, the
 ↪  exponential term dominates and captures the leading-order behavior, while the
 ↪  prefactor varies more slowly. In stochastic systems, the small parameter \(
 ↪  \varepsilon \) appears in the exponent, leading to sharp concentration of the
 ↪  probability distribution around stable points as \( \varepsilon \to 0 \). The
 ↪  quasi-potential \( V(x) \) acts as an effective potential or energy landscape,
 ↪  quantifying the cost of rare fluctuations. The symbol ``\( \sim \)'' denotes
 ↪  logarithmic equivalence, meaning that
 \[
-\lim_{\varepsilon \to 0} \frac{\log p_s(x)}{-\varepsilon^{-1} V(x)} = 1,
-\]
-which highlights that \( V(x) \) governs the leading-order behavior in the small noise
↪  limit.
-
-The concept of the quasi-potential originates from large deviation theory
↪  \cite{Freidlin2012}. It is defined via a variational principle involving the
↪  Freidlin-Wentzell action functional \( S_T[\phi] \), formulated over continuous
↪  trajectories \( \phi(t) \) that connect a stable fixed point \( \bar{x} \) to a
↪  target point \( x \). Specifically, the quasi-potential is given by the nested
↪  infimum:
+\lim_{\varepsilon \to 0}-\varepsilon\log p_s(x)=V(x),
+\]
+after compatible normalization and in the relevant single-attractor asymptotic
↪  setting. In a multistable problem, the stationary rate function and the action
↪  value relative to one attractor must be distinguished.
+
+The concept of the quasi-potential originates from large deviation theory
↪  \cite{Freidlin2012}. It is defined via a variational principle involving the
↪  Freidlin--Wentzell action functional \( S_T[\phi] \), formulated over continuous
↪  trajectories \( \phi(t) \) that connect a stable fixed point \( \bar{x} \) to a
↪  target point \( x \). Specifically, the quasi-potential is given by the nested
↪  infimum:
 \begin{equation}
 \label{quasip}
-V(x) \equiv \inf_{T > 0} \inf_{\phi \in C([0, T])} \left\{ S_T[\phi] \,:\, \phi(0) =
↪  \bar{x},\; \phi(T) = x \right\},
-\end{equation}
-where \( C([0,T]) \) denotes the space of continuous functions on the interval \([0,
↪  T]\). The action functional \( S_T[\phi] \), fundamental in the large deviation
↪  framework, is given by
+V(x) \equiv \inf_{T > 0} \inf_{\phi \in AC([0, T])} \left\{ S_T[\phi] \,:\, \phi(0) =
↪  \bar{x},\; \phi(T) = x \right\},
+\end{equation}
+where \(AC([0,T])\) denotes absolutely continuous paths; the action is infinite
↪  outside its admissible path class. The action functional \( S_T[\phi] \),
↪  fundamental in the large deviation framework, is given by
 \begin{equation}
 \label{actionf}
 S_T[\phi] = \int_0^T L(\phi(t), \dot{\phi}(t))\,dt,
@@ -146,16 +145,16 @@
 \end{equation}
 for small \( \delta > 0 \), illustrating that the action functional governs the
 ↪  exponential rate of decay of such pathwise probabilities.
```

```
-Note that the stationary probability distribution satisfies the time-independent
↪  Fokker-Planck equation
-\[ \mathcal{L}p_s(x) = - \sum_{i=1}^n \frac{\partial}{\partial x_i} b_i(x)p_{s}(x) +
↪  \sum_{i,j=1}^n \frac{\partial^2}{\partial x_i \partial x_j} a_{ij}(x)p_{s}(x) = 0,
↪  \]
-where \( \mathcal{L} \) is the Fokker-Planck operator. Substituting the WKB ansatz
↪  \eqref{WKB} into this equation and collecting the leading-order terms in \(
↪  \varepsilon \) yields the Hamilton-Jacobi equation:
+Note that the stationary probability distribution satisfies the time-independent
↪  Fokker--Planck equation
+\[ \mathcal{L}p_s(x) = - \sum_{i=1}^n \frac{\partial}{\partial
↪  x_i}\bigl(b_i(x)p_{s}(x)\bigr) + \frac{\varepsilon}{2}\sum_{i,j=1}^n
↪  \frac{\partial^2}{\partial x_i \partial x_j}\bigl(a_{ij}(x)p_{s}(x)\bigr) = 0, \]
+where \( \mathcal{L} \) is the Fokker--Planck operator. Substituting the WKB ansatz
↪  \eqref{WKB} into this equation and collecting the leading-order terms in \(
↪  \varepsilon \) yields the Hamilton--Jacobi equation:
 \begin{equation}
 \label{HJE}
 H(x, p) \equiv b^{T}p + \frac{1}{2} p^{T}ap = 0, \quad p \equiv \nabla V (x),
 \end{equation}
 which is a first-order nonlinear partial differential equation for the quasi-potential
 ↪  \( V \). It is observed that this Hamiltonian is the Legendre transform of the
 ↪  Lagrangian \eqref{Lagran}. According to the definition \eqref{quasip}, the
 ↪  quasi-potential at the fixed point satisfies \( V(\bar{x}) = 0 \). Since the
 ↪  quasi-potential function attains minimal value at the stable fixed point, the
 ↪  gradient of the quasi-potential is also zero at the stable point, i.e., $\nabla
 ↪  V(\bar{x}) =0$.

-If \( b(x) = -\nabla U(x) \) (i.e., a gradient system) and \( a(x) = I \) (i.e.,
↪  additive noise), then the quasi-potential function equals twice the potential: \(
↪  V(x) = 2U(x) \). In this special case, the quasi-potential admits an analytical
↪  expression.
+If \( b(x) = -\nabla U(x) \) (i.e., a gradient system) and \( a(x) = I \) (i.e.,
↪  additive noise), then within the relevant attraction basin the quasi-potential is
↪  \(V(x)=2[U(x)-U(\bar x)]\), under the usual connecting-path assumptions. In this
↪  special case, the quasi-potential admits an analytical expression.

 Assume that the second-order approximation of the quasi-potential function at the
 ↪  fixed point \( \bar{x} \) is given by
 \[
@@ -168,7 +167,7 @@
 \[
 \nabla V(x) \approx Z (x - \bar{x}).
 \]
-Substituting the gradient approximation into the Hamilton-Jacobi equation (\ref{HJE})
↪  yields
+Substituting the gradient approximation into the Hamilton--Jacobi equation (\ref{HJE})
↪  yields
 \begin{equation}\label{HJE2}
 b(x)^T Z (x - \bar{x}) + \frac{1}{2} (x - \bar{x})^T Z^T a(x) Z (x - \bar{x}) \approx
 ↪  0.
 \end{equation}
@@ -184,7 +183,7 @@
 \begin{equation}\label{HJE4}
 (x - \bar{x})^T (F^TZ + ZF + Z^TAZ) (x - \bar{x}) \approx 0.
 \end{equation}
```

```
-\textcolor{red}{Since the matrix $F^TZ + ZF + Z^TAZ$ is a symmetric matrix, it must be
↪  diagonalizable. Due to the arbitrariness of $x - \bar{x}$, the eigenvalues of the
↪  matrix must all be 0. That is, each element of the matrix is 0.} Therefore, the
↪  Hessian matrix \( Z = \nabla^2 V(\bar{x}) \) satisfies the algebraic Riccati
↪  equation:
+Since the matrix $F^TZ + ZF + Z^TAZ$ is a symmetric matrix, it must be diagonalizable.
↪  Due to the arbitrariness of $x - \bar{x}$, the eigenvalues of the matrix must all
↪  be 0. That is, each element of the matrix is 0. Therefore, the Hessian matrix \( Z
↪  = \nabla^2 V(\bar{x}) \) satisfies the algebraic Riccati equation:
 \begin{equation}\label{HJE5}
  F^TZ + ZF + Z^TAZ= 0
 \end{equation}
@@ -196,44 +195,42 @@

 The principal challenge in computing the quasi-potential from the PDE \eqref{HJE} lies
 ↪  in avoiding trivial solutions. To address this, we employ kernel methods to solve
 ↪  the PDE \eqref{HJE}, and use equation \eqref{Riccati} to impose that the Hessian
 ↪  of \( V \) at the origin coincides with the matrix \( Z \) solving
 ↪  \eqref{Riccati}.

-It is worth emphasizing that the quasi-potential plays a pivotal role not only in
↪  large deviation theory but also in stochastic dynamics, as it is linked to various
↪  physical quantities. For example, the expected time to exit the domain of
↪  attraction of a fixed point-known as the mean exit time-is exponentially determined
↪  by the minimum of the quasi-potential on the boundary of the domain. The exit
↪  location distribution and the invariant measure can also be approximated based on
↪  the quasi-potential. Furthermore, once the quasi-potential function is known, the
↪  most probable transition path connecting two steady states can be derived from the
↪  quasi-potential. Typically, the most probable path passes through a saddle point,
↪  with the latter half of the trajectory beyond the saddle point following its
↪  unstable manifold toward the destination steady state. The first half of the path
↪  is obtained by solving the equation \begin{equation}\label{mppcomp}
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↪  large deviation theory but also in stochastic dynamics, as it is linked to various
↪  physical quantities. For example, the expected time to exit the domain of
↪  attraction of a fixed point---known as the mean exit time---is exponentially
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↪  point, with the latter half of the trajectory beyond the saddle point following its
↪  unstable manifold toward the destination steady state. The first half of the path
↪  is obtained by solving the equation \begin{equation}\label{mppcomp}
 \dot{x} = b(x) + a(x) \nabla V(x).
 \end{equation}
-The boundary conditions are $x(-\infty)=\bar{x}$ and $x(\infty)=x_{sa}$, where
↪  $x_{sa}$ denotes the saddle point. In practice, Equation (\ref{mppcomp}) is
↪  obtained by substituting $p= \nabla V(x)$ into the state equation $\dot{x} = b(x) +
↪  a(x)p$ of the Hamiltonian system corresponding to the Hamilton-Jacobi Equation
↪  (\ref{HJE}). Therefore, the most probable path is the projection of a heteroclinic
↪  orbit in the four-dimensional Hamiltonian phase space onto the two-dimensional
↪  space. For further properties of the quasi-potential, see \cite{Freidlin2012}.
↪  \begin{remark}
```

```
+The boundary conditions are $x(-\infty)=\bar{x}$ and $x(\infty)=x_{sa}$, where
↪  $x_{sa}$ denotes the saddle point. In practice, Equation (\ref{mppcomp}) is
↪  obtained by substituting $p= \nabla V(x)$ into the state equation $\dot{x} = b(x) +
↪  a(x)p$ of the Hamiltonian system corresponding to the Hamilton-Jacobi Equation
↪  (\ref{HJE}). Therefore, the most probable path is the projection of a heteroclinic
↪  orbit in the $2n$-dimensional Hamiltonian phase space onto the $n$-dimensional
↪  state space. For further properties of the quasi-potential, see
↪  \cite{Freidlin2012}. \begin{remark}
 When $\varepsilon = 0$ (i.e. when there is no noise), our work will reduce to
 ↪  constructing the Lyapunov function for the unforced system $\dot{x} = b(x)$
 ↪  \cite{giesl2019approximation, lee2024note}
 \end{remark}


 \section{Method for Approximating the Quasi-Potential}\label{methsec}

-To approximate the quasi-potential, we exploit the spectral structure of the Koopman
↪  operator. Specifically, we construct eigenfunctions \( \varphi_i(x) \) satisfying
↪  the linear advection-type partial differential equations:
-\begin{equation}
-\nabla \varphi_i(x) \cdot b(x) = \lambda_i \varphi_i(x), \quad \varphi_i(\bar{x}) = 0,
↪  \quad \nabla \varphi_i(\bar{x}) = e_i,
-\end{equation}
-where \( \lambda_i \) are eigenvalues of the linearization \( J = \nabla b(\bar{x}) \)
↪  at the fixed point \( \bar{x} \), and \( e_i \) are the standard basis vectors.
↪  These eigenfunctions correspond to the Koopman eigenfunctions associated with the
↪  equilibrium dynamics and encode local geometric and dynamical information such as
↪  the structure of stable and unstable manifolds.
-
-Let \( \Phi(x) = (\varphi_1(x) \cdots \varphi_n(x))^T \), and define the matrix \( W
↪  \) whose columns are the eigenvectors \( w_i \) of \( J \), satisfying \( J w_i =
↪  \lambda_i w_i \). We then define a quadratic approximation to the quasi-potential:
-\begin{equation} \label{eq:vmid}
-V_{\text{mid}}(x) = \tfrac{1}{2} \Phi(x)^T P_{\text{mid}} \Phi(x),
-\end{equation}
-with the symmetric matrix\footnote{Alternatively, one can set \( P_{\text{mid}} =
↪  \frac{1}{2}(P_u + P_\ell) \), where \( P_u \) and \( P_\ell \) solve Riccati-type
↪  equations: \( P \Lambda + \Lambda^\top P + PRP = 0 \), for different choices of \(
↪  R \).}:
-\begin{equation}
-P_{\text{mid}} = W\,\mathrm{diag}(-2\lambda_i)\,W^{-1}.
-\end{equation}
+We consider both direct scalar kernel approximations and a structured spectral
↪  representation. The latter exploits the Koopman operator; its construction is
↪  developed first, and the direct finite-feature implementation is given in
↪  Section~\ref{sec:direct-implementation}. Specifically, we construct eigenfunctions
↪  \( \varphi_i(x) \) satisfying the linear advection-type partial differential
↪  equations:
+\begin{equation}
+\nabla \varphi_i(x) \cdot b(x) = \lambda_i \varphi_i(x), \quad \varphi_i(\bar{x}) = 0,
↪  \quad \nabla \varphi_i(\bar{x}) = \ell_i,
+\end{equation}
```

```
+where \( \lambda_i \) are eigenvalues of the linearization \( J = \nabla b(\bar{x}) \)
↪  at the fixed point \( \bar{x} \), and \(\ell_i\) are left eigenvectors,
↪  \(J^\top\ell_i=\lambda_i\ell_i\). Standard basis vectors give this normalization
↪  only in suitable coordinates. Complex conjugate modes are handled through their
↪  real and imaginary parts; smooth independent eigenfunctions on an entire basin
↪  require additional dynamical assumptions. These eigenfunctions correspond to the
↪  Koopman eigenfunctions associated with the equilibrium dynamics and encode local
↪  geometric and dynamical information such as the structure of stable and unstable
↪  manifolds.
+
+Let $\Phi=(\varphi_1,\ldots,\varphi_n)^\top$ in a real coordinate representation, and
↪  let $S$ be the invertible matrix of prescribed gradients at $\bar x$ (rows
↪  $\ell_i^\top$, or corresponding real blocks). We define
+\begin{equation}\label{eq:vmid}
+ V_{\mathrm{mid}}(x)=\tfrac12\Phi(x)^\top P_{\mathrm{mid}}\Phi(x),
+ \qquad P_{\mathrm{mid}}=S^{-\top}P_{\mathrm{loc}}S^{-1},
+\end{equation}
+where $P_{\mathrm{loc}}$ is the positive-definite physical Riccati Hessian. This
↪  matrix is symmetric and includes the diffusion information. If the value and
↪  gradient normalizations of $\Phi$ hold exactly, the Hessian of $V_{\rm mid}$ is
↪  exactly $P_{\rm loc}$; the curvature penalty remains useful for approximate
↪  enforcement. A similarity transform of eigenvalues alone is not a general formula
↪  for this Hessian.

 To represent \( \varphi_i(x) \), we employ a data-driven kernel-based ansatz:
 \begin{equation}\label{phiix}
-\varphi_i(x) = \sum_{p=1}^N \alpha_p^{(i)}\,k_\beta(x, x_p) + (x - \bar{x}) \cdot w_i,
-\end{equation}
-where the point set \( \{ x_p \}_{p=1}^N \) denotes the collocation set which is used
↪  for training, and \( k_\beta(x, t) \) is a convex combination of base kernels
↪  (e.g., polynomials and radial basis functions), parameterized by a non-negative
↪  weight vector \( \beta \). This kernel formulation ensures flexibility and
↪  regularization, and it circumvents the need to construct the Koopman operator
↪  explicitly.
-
-The coefficients $\alpha_p$ are determined by minimizing the regularized empirical
↪  loss functional using the representer theorem.
+\varphi_i(x) = \sum_{p=1}^N \alpha_p^{(i)}\,k_\beta(x, x_p) + (x - \bar{x}) \cdot
↪  \ell_i,
+\end{equation}
+where the point set \( \{ x_p \}_{p=1}^N \) denotes the collocation set which is used
↪  for training, and \( k_\beta(x, t) \) is $k_\beta=\sum_j\beta_j^2k_j$, a convex
↪  combination of base kernels with nonnegative amplitudes $\beta$ satisfying
↪  $\sum_j\beta_j^2=1$. This kernel formulation ensures flexibility and
↪  regularization, and it circumvents the need to construct the Koopman operator
↪  explicitly.
+
+The coefficients $\alpha_p$ are determined by a regularized finite-section collocation
↪  fit. A derivative-loss representer theorem involves the corresponding derivative
↪  functionals and does not by itself identify the quadratic eigenfunction surrogate
↪  with a scalar RKHS minimizer.
 To determine the coefficients $\alpha^{(i)} = (\alpha^{(i)}_1,\dots,\alpha^{(i)}_N)$,
 we enforce the Koopman eigenfunction equation at the collocation points.
 Specifically, for each $i=1,\dots,n$, we impose the conditions
 \[
 \varphi_i(\bar x)=0, \qquad
-\nabla\varphi_i(\bar x)=e_i,
+\nabla\varphi_i(\bar x)=\ell_i,
 \]
```

```
 together with the PDE constraint
 \[
@@ -263,243 +260,148 @@
 $\alpha^{(i)}$ used to construct $\Phi(x)$ and the quasi-potential
 surrogate $V_{\rm mid}(x)$.

-\textcolor{red}{
+
 \begin{remark}[RKHS norm and the linear correction term]
-The ansatz~\eqref{phiix} decomposes $\varphi_i$ into an RKHS part $\sum_p
↪  \alpha_p^{(i)} k_\beta(x,x_p)$ and a fixed linear term $(x-\bar{x})\cdot w_i$. The
↪  RKHS norm $\|\varphi_i\|_{\mathcal{H}_k}^2 = \alpha^{(i)T}G_\beta\alpha^{(i)}$
↪  penalizes only the learnable kernel component, not the linear correction. This is
↪  intentional: the linear term is analytically prescribed by the boundary condition
↪  $\nabla\varphi_i(\bar{x})=e_i$ and carries zero free parameters, so it should not
↪  be regularized. Penalizing it would bias the solution away from the exact local
↪  linearization.
+The ansatz~\eqref{phiix} decomposes $\varphi_i$ into an RKHS part $\sum_p
↪  \alpha_p^{(i)} k_\beta(x,x_p)$ and a fixed linear term $(x-\bar{x})\cdot\ell_i$.
↪  The RKHS norm $\|\varphi_i\|_{\mathcal{H}_k}^2 = \alpha^{(i)T}G_\beta\alpha^{(i)}$
↪  penalizes only the learnable kernel component, not the linear correction. This is
↪  intentional: the linear term is analytically prescribed by the boundary condition
↪  $\nabla\varphi_i(\bar{x})=\ell_i$ and carries zero free parameters, so it should
↪  not be regularized. Penalizing it would bias the solution away from the exact local
↪  linearization.
 \end{remark}
-}
-
-To determine optimal kernel weights \( \beta \), we minimize the following regularized
↪  loss function, \textcolor{red}{where $P_{\text{loc}} \in \mathbb{R}^{n \times n}$
↪  denotes the local curvature target matrix---specifically, the positive-definite
↪  solution of the algebraic Riccati equation~\eqref{Riccati} evaluated at $\bar{x}$:}
+
+
+To determine optimal kernel weights \( \beta \), we minimize the following regularized
↪  loss function, where $P_{\text{loc}} \in \mathbb{R}^{n \times n}$ denotes the local
↪  curvature target matrix---specifically, the positive-definite solution of the
↪  algebraic Riccati equation~\eqref{Riccati} evaluated at $\bar{x}$:

 \begin{align}\label{eq:composite_loss}
 L(\beta) = & \underbrace{\sum_p \left[ b^{T}(x_p)\nabla V_{\text{mid}}(x_p) +
 ↪  \frac{1}{2} \nabla V_{\text{mid}}(x_p)^{T}a(x_p)\nabla
 ↪  V_{\text{mid}}(x_p)\right]^2}_{L_{\text{HJ}}}}
 + \gamma \underbrace{\|\nabla^2 V_{\text{mid}}(\bar{x}) -
 ↪  P_{\text{loc}}\|_F^2}_{L_{\text{curv}}} \notag \\
 & + \lambda_1 \sum_j |\beta_j| + \eta \sum_i \|\varphi_i\|_{\mathcal{H}_k}^2,
 \end{align}
-subject to the constraints \( \beta_j \ge 0 \) and \( \sum_j \beta_j = 1 \). Here $\|
↪  \cdot \|_F$ denotes the Frobenius norm. \textcolor{red}{The first term
↪  $L_{\text{HJ}}$ enforces consistency with the Hamilton--Jacobi structure; the
↪  second term $L_{\text{curv}}$ aligns the local curvature of the approximation with
↪  that of the Riccati solution at $\bar{x}$. The remaining terms promote sparsity and
↪  RKHS regularity.}
+subject to the constraints \( \beta_j \ge 0 \) and \( \sum_j \beta_j^2 = 1 \). Here
↪  $\| \cdot \|_F$ denotes the Frobenius norm. The first term $L_{\text{HJ}}$ enforces
↪  consistency with the Hamilton--Jacobi structure; the second term $L_{\text{curv}}$
↪  aligns the local curvature of the approximation with that of the Riccati solution
↪  at $\bar{x}$. The remaining terms promote sparsity and RKHS regularity.
+
```

```
+\paragraph{Kernel amplitudes versus mixture weights.}\label{sec:amplitudes}
+We use nonnegative amplitudes $\beta_j$ with $\sum_j\beta_j^2=1$ and define the actual
↪  mixture weights as $w_j=\beta_j^2$. Thus $k_\beta=\sum_j\beta_j^2k_j$ is a convex
↪  kernel combination, while the sparsity penalty is applied to the amplitudes. It is
↪  nonconstant:
+\[
+ 1\leq\sum_j\beta_j\leq\sqrt L.
+\]
+Its minimum is attained by a single active amplitude, so it encourages concentrated
↪  mixtures; it does not guarantee that a nonconvex optimizer finds a particular
↪  sparse model. In contrast, an $\ell_1$ penalty on simplex weights $w_j\geq0$,
↪  $\sum_jw_j=1$ is constant and has no sparsifying effect. The distinction is
↪  essential. Pruning small weights or amplitudes is an additional heuristic, followed
↪  by renormalization and refitting. In a single-component model the active amplitude
↪  and its weight both equal one. The discrete polynomial-degree selection used in
↪  Section~\ref{examples} is specified separately from continuous mixture
↪  optimization.

 \paragraph{Algorithmic structure.}
-In practice, the computation proceeds in two stages. For a fixed choice of kernel
↪  weights $\beta$, the coefficients $\alpha^{(i)}$ are obtained by solving the
↪  regularised linear systems (\ref{eqn:alpha_eqn}), yielding the Koopman
↪  eigenfunction approximations $\varphi_i$ and hence $\Phi(x)$.  In a second, outer
↪  optimisation loop, the kernel mixture coefficients $\beta$ are updated by
↪  minimising the composite loss (\ref{eq:composite_loss}), while at each iteration
↪  the functions $\varphi_i$ are recomputed from the updated kernel $k_\beta$.  This
↪  bi-level structure (inner linear solves for $\alpha^{(i)}$, outer nonlinear
↪  optimisation for $\beta$) matches the implementation used in the numerical
↪  experiments of Section \ref{examples}.
-
-\textcolor{red}{
+In practice, the computation proceeds in two stages. For a fixed choice of kernel
↪  weights $\beta$, the coefficients $\alpha^{(i)}$ are obtained by solving the
↪  regularised linear systems (\ref{eqn:alpha_eqn}), yielding the Koopman
↪  eigenfunction approximations $\varphi_i$ and hence $\Phi(x)$.  In a second, outer
↪  optimisation loop, the kernel mixture coefficients $\beta$ are updated by
↪  minimising the composite loss (\ref{eq:composite_loss}), while at each iteration
↪  the functions $\varphi_i$ are recomputed from the updated kernel $k_\beta$.  This
↪  bi-level structure separates the inner transport objective from the outer HJ
↪  objective. The inner and outer objectives play different roles when comparing the
↪  hybrid construction with a direct scalar fit.
+
+
 \paragraph{Kernel dictionary.}
```

```
-Throughout our numerical experiments, we employ a hybrid dictionary combining Gaussian
↪  radial basis function (RBF) kernels and polynomial kernels. This choice is
↪  motivated by complementary approximation properties: RBF kernels excel at local
↪  interpolation and capturing rapid variations near critical points, while polynomial
↪  kernels naturally encode global algebraic structure characteristic of
↪  quasi-potentials. In our previous work \cite{hamzi2021learning}, this
↪  RBF+polynomial combination demonstrated robust performance across diverse dynamical
↪  systems benchmarks. The $\ell_1$ sparsity penalty in the composite
↪  loss~\eqref{eq:composite_loss} acts as an automatic model selection mechanism
↪  analogous to MKL \cite{lanckriet2004learning, bach2004multiple}: kernels that do
↪  not contribute meaningfully to the approximation have their weights driven to zero
↪  during optimization. In principle, the framework can accommodate much larger and
↪  more diverse kernel dictionaries; in \cite{bh_sparse_kfs} the same $\ell_1$-based
↪  sparsification mechanism was used with an extensive kernel library to learn 132
↪  chaotic dynamical systems. The methodology naturally extends to other kernel
↪  families (e.g., Mat\'ern, rational quadratic, or domain-specific kernels) when more
↪  complex functional structures are anticipated.
-}
-
-\paragraph{Theoretical guarantee.}
-The composite loss \eqref{eq:composite_loss} couples three ingredients:\begin{itemize}
-    \item[i.] a Hamilton--Jacobi residual that enforces the global large-deviation
-structure, \item[ii.]  an $\ell_{1}$ / RKHS regulariser that controls capacity,
-and \item[iii.]  a Riccati-curvature term that anchors the surrogate to the exact
-quadratic well at the attractor.
-\end{itemize}
-  The next result shows that, under the
-standard assumptions of kernel-ridge learning and an appropriate schedule
-for the regularisation weights, these ingredients are sufficient to drive
-the surrogate \(V_{N}\) to the true quasi-potential \(V\) as the number of
-collocation points \(N\) grows.  In particular, the Hessian of \(V_{N}\)
-at the equilibrium converges to the Riccati matrix \(P\), giving a
-second-order match that accelerates the finite-sample error decay observed
-in Section~\ref{examples}.  We state the result and the proof for completeness.
-
-
-\begin{theorem}[Consistency with Riccati-curvature
↪  matching]\label{thm:consistency-curvature}
-Let $V:\Omega\subset\mathbb R^{d}\to\mathbb R_{\ge0}$ be the unique viscosity solution
↪  of the stationary Hamilton--Jacobi equation
-\[
-  H\bigl(x,\nabla V(x)\bigr)=0,\qquad V(\bar x)=0,
-\]
-and let $P\succ0$ be the Riccati matrix
-\[
-  b'(\bar x)^{\!\top}P + P\,b'(\bar x) \;=\; P\,a(\bar x)\,P.
-\]
-
-\medskip\noindent
-Assume:
```

```
+The general framework admits a dictionary combining Gaussian radial basis function
↪  (RBF) kernels and polynomial kernels. Section~\ref{examples} examines polynomial
↪  components, and Appendix~\ref{app:additional} evaluates Gaussian kernels. This
↪  choice is motivated by complementary approximation properties: RBF kernels excel at
↪  local interpolation and capturing rapid variations near critical points, while
↪  polynomial kernels naturally encode global algebraic structure characteristic of
↪  quasi-potentials. RBF+polynomial dictionaries have demonstrated robust performance
↪  across diverse dynamical-systems benchmarks \cite{hamzi2021learning}. Applied to
↪  the normalized amplitudes specified above, the $\ell_1$ term in the composite
↪  loss~\eqref{eq:composite_loss} encourages concentrated kernel mixtures, in the
↪  spirit of MKL \cite{lanckriet2004learning, bach2004multiple}: small components may
↪  be suppressed during fitting and removed by the stated pruning rule; exact sparse
↪  recovery is not guaranteed. In principle, the framework can accommodate much larger
↪  and more diverse kernel dictionaries; in \cite{bh_sparse_kfs} the same
↪  $\ell_1$-based sparsification mechanism was used with an extensive kernel library
↪  to learn 132 chaotic dynamical systems. The methodology naturally extends to other
↪  kernel families (e.g., Mat\'ern, rational quadratic, or domain-specific kernels)
↪  when more complex functional structures are anticipated.
+
+
+\paragraph{Role and limitations of the Koopman
↪  representation.}\label{sec:koopman-role}
+A direct comparator is $V_D(x)=\tfrac12(x-\bar x)^\top P_{\rm loc}(x-\bar x)+f_k(x)$,
↪  where the kernel correction can be constrained to have zero value, gradient, and
↪  Hessian at $\bar x$. Such a scalar model can minimize the same HJ residual without
↪  first learning eigenfunctions. Both methods permit analytic differentiation. The
↪  spectral model instead fits $n$ transport equations, provides nonnegativity when
↪  $P_{\rm mid}\succ0$, and supplies dynamical coordinates that may be reused in other
↪  tasks. These are structural advantages, not a proof of better accuracy or lower
↪  total cost.
+
+In real coordinates with $D\Phi(\bar x)=I$, suppose $D\Phi\,b=F\Phi$ exactly and set
↪  $V_K=\tfrac12\Phi^\top P\Phi$. Then
+\begin{equation}\label{eq:transformed-diffusion}
+ H(x,\nabla V_K)=\tfrac12\Phi^\top
+ \bigl(F^\top P+PF+P a_\Phi(x)P\bigr)\Phi,
+ \quad a_\Phi=D\Phi\,a\,D\Phi^\top.
+\end{equation}
+The transformed diffusion is generally state dependent, so local Riccati matching does
↪  not make this expression vanish everywhere. For example, $b(x)=-x(1+x^2)$,
↪  $a(x)=1+x^2$ has $V(x)=x^2$ and the normalized eigenfunction
↪  $\varphi=x/\sqrt{1+x^2}$ with eigenvalue $-1$. The quadratic spectral candidate is
↪  $V_K=\varphi^2=x^2/(1+x^2)$, with
+\[
+ H(x,V_K')=-\frac{2x^4(2+x^2)}{(1+x^2)^3}.
+\]
+It matches the correct Hessian at zero but is not the quasi-potential. Thus a finite
↪  quadratic eigenfunction representation can have approximation bias even with exact
↪  eigenfunctions. Appendix~\ref{app:additional} compares a direct scalar correction
↪  and this restricted spectral construction using common data and curvature
↪  constraints; the tests favor the direct construction within these specified model
↪  classes. The Gaussian-kernel models in that comparison complement the
↪  polynomial-kernel experiments in Section~\ref{examples}.
+
+\paragraph{Theoretical guarantee and solution
↪  selection.}\label{sec:solution-selection}
```

```
+The composite loss \eqref{eq:composite_loss} couples a Hamilton--Jacobi residual, a
↪  regularizer, and a Riccati-curvature term. These ingredients address PDE fit,
↪  capacity, and local geometry, respectively. A convergence statement must also
↪  identify the action-selected solution. Strict convexity of $H(x,\cdot)$ does not
↪  supply this selection: for $b(x)=-x$ and $a=1$, both $u=0$ and $u=x^2$ solve the HJ
↪  equation and satisfy $u(0)=0$. Curvature matching excludes the constant candidate
↪  locally but does not by itself establish global uniqueness. We therefore state the
↪  comparison, compactness, and residual-control requirements explicitly, following
↪  viscosity stability \cite{Crandall1992}.
+
+\begin{theorem}[Conditional consistency with Riccati-curvature
↪  matching]\label{thm:consistency-curvature}
+Let $D\subset\mathbb R^d$ be a bounded domain with compact closure. Suppose $b,a$ are
↪  continuous, $a$ is uniformly positive definite, and compatible continuous boundary
↪  data $g$ select the desired restriction $V$ of the quasi-potential through
+\[
+H(x,\nabla V)=0\quad\text{in }D,\qquad V=g\quad\text{on }\partial D.
+\]
+Assume that this boundary-value problem has a continuous viscosity solution and a
↪  comparison principle in the relevant bounded continuous class. Let $V_N\in
↪  C^1(\overline D)$ be a sequence of kernel-based surrogates satisfying:
 \begin{enumerate}
-  \item\label{A1} $b,a\in C^{1}(\Omega)$ and $a(x)$ is uniformly positive definite
↪  on~$\Omega$;
-  \item\label{A2} $k$ is a continuous, $C^{m}$ $(m\!\ge2)$ \emph{universal} kernel
↪  on~$\Omega$ with RKHS $\mathcal H_{k}$;
-  \item\label{A3} Collocation points $\{x_{p}\}_{p=1}^{N}$ are drawn i.i.d.\ from a
↪  probability measure $\rho$ whose support is $\overline\Omega$;
-  \item\label{A4} Regularisation parameters follow
-        $\lambda_{N}\!\downarrow\!0$, $N\lambda_{N}\!\uparrow\!\infty$,
-        and a curvature weight $\gamma_{N}$ satisfies
-        $\gamma_{N}\!\downarrow\!0$ with $\gamma_{N}/\lambda_{N}=O(1)$.
+\item $V_N$ are uniformly bounded and equicontinuous on $\overline D$;
+\item $\|V_N-g\|_{L^\infty(\partial D)}\to0$;
+\item for $r_N(x)=H(x,\nabla V_N(x))$, the collocation sets $X_N$ have fill distance
↪  $h_N\to0$, the residuals have a common modulus of continuity $\omega$, and
↪  $\max_{x_p\in X_N}|r_N(x_p)|\to0$.
 \end{enumerate}
-
-\medskip\noindent
-For $N\in\mathbb N$, define the empirical objective:
+Then $V_N\to V$ uniformly on $\overline D$, and hence in $L^2(\rho)$ for every
↪  probability measure $\rho$ on that closure.
+
+If, in addition, the functions are $C^2$ near $\bar x$ and satisfy
+$\|D^2V_N(\bar x)-P\|_F\to0$, their equilibrium curvature converges to $P$, where
+\[
+ b'(\bar x)^\top P+P b'(\bar x)+P a(\bar x)P=0,\qquad P\succ0.
+\]
+Gradient convergence requires additional compactness: if $V\in C^1$ and $\{\nabla
↪  V_N\}$ is relatively compact in $C(K)$ on every compact $K\Subset D$, then $\nabla
↪  V_N\to\nabla V$ uniformly on such $K$. None of these compactness or
↪  solution-selection conditions follows solely from small empirical loss.
+\end{theorem}
+
+\begin{remark}[Notation and empirical objective]
```

```
+$V_N$ denotes a sample-dependent approximation, $V_{\rm mid}$ the quadratic Koopman
↪  surrogate, and $V_h$ or $V_{\rm quad}$ particular numerical reconstructions. Write
↪  $\mathcal R(\Phi,\beta)$ for the nonnegative regularizer of the learnable kernel
↪  components and amplitudes. The normalized empirical objective is
 \begin{equation}\label{eq:empirical-loss}
-  \mathcal L_{N,\lambda,\gamma}(\Phi,\beta)
-  := \frac1N\sum_{p=1}^{N} %
-     H\!\bigl(x_{p},\nabla V_{(\Phi,\beta)}(x_{p})\bigr)^{2}
-     \;+\;
-     \gamma\,\!\bigl\|D^{2}V_{(\Phi,\beta)}(\bar x)-P\bigr\|_{F}^{2}
-     \;+\;
-     \lambda\,\!\bigl(\|\Phi\|^{2}_{\mathcal H_{k}} + \|\beta\|_{1}\bigr),
-\end{equation}
-and let $(\widehat\Phi_{N},\widehat\beta_{N})$ be any minimiser
↪  of~\eqref{eq:empirical-loss}.
-Set $V_{N}:=V_{(\widehat\Phi_{N},\widehat\beta_{N})}$.
-
-\medskip\noindent
-Then, as $N\to\infty$,
-\[
-  \|V_{N}-V\|_{L^{2}(\rho)} \;\xrightarrow{\;p\;} 0,
-  \qquad
-  \|\nabla V_{N}-\nabla V\|_{L^{2}(\rho)}
-     \;\xrightarrow{\;p\;} 0,
-  \qquad
-  D^{2}V_{N}(\bar x)\;\xrightarrow{\;p\;} P.
-\]
-If $V\in C^{m}(\Omega)$, the convergence is uniform on compact subsets of~$\Omega$.
-\end{theorem}
-
-% ADD after the proof of Theorem 4.1:
-
-\begin{remark}[Notation]
-In the theoretical analysis (Theorem \ref{thm:consistency-curvature}), we use $V_N$ to
↪  denote the $N$-sample approximation obtained by minimizing the empirical objective.
↪  In the numerical examples (Section~\ref{examples}), we use $V_{\text{mid}}$ to
↪  denote the mid-level Koopman-based approximation constructed via
↪  Equation~\eqref{eq:vmid}, and $V_h$ or $V_{\text{quad}}$ for specific baseline or
↪  hybrid approximations. These notational variants all refer to finite-sample
↪  approximations of the true quasi-potential $V$, distinguished only by their
↪  construction method or the context in which they appear.
+\begin{aligned}
+\mathcal L_{N,\lambda,\gamma}(\Phi,\beta)
++={}&\frac1N\sum_{p=1}^N H(x_p,\nabla V_{(\Phi,\beta)}(x_p))^2\\
+&+\gamma\|D^2V_{(\Phi,\beta)}(\bar x)-P\|_F^2
++\lambda\mathcal R(\Phi,\beta).
+\end{aligned}
+\end{equation}
+The theorem concerns sequences meeting its stated conditions, not arbitrary local or
↪  global minimizers of this nonconvex objective. Its boundary data are an additional
↪  solution-selection condition, not the single point constraint $V_N(\bar x)=0$ used
↪  in a local fit.
 \end{remark}

-\begin{proof}[\textcolor{red}{Proof of Theorem~\ref{thm:consistency-curvature}}]
-We follow the classical excess-risk decomposition of kernel ridge regression
↪  \cite{CaponnettoDeVito2007,SteinwartChristmann2008}, extended with curvature
↪  matching.
-
-\medskip
```

```
-\noindent\textbf{Step 1: Generalized representer theorem.}
-The empirical loss in~\eqref{eq:empirical-loss} depends on three types of evaluations:
-function values and gradients at the collocation points, and second derivatives at a
↪  single fixed point $\bar x$.
-By the generalized representer theorem, this structure implies that $V_N$ admits a
↪  finite-dimensional representation:
-
-\[
-  V_{N}(x)
-  = \sum_{p=1}^{N}\!\sum_{j=1}^{n}
-        c_{pj}\,\partial_{x_{j}}k(x,x_{p})
-    + \sum_{1\le j\le\ell\le n}
-        d_{j\ell}\,\partial_{x_{j}x_{\ell}}k(x,\bar x),
-\]
-with coefficients $\{c_{pj}\},\{d_{j\ell}\}$ determined by the optimiser. All
↪  derivatives belong to $\mathcal H_{k}$ by the $C^m$ assumption on $k$, and the RKHS
↪  norm is bounded by
-\[
-  \|V_{N}\|_{\mathcal H_{k}}
-  \;\le\;
-  C\|\theta_{N}\|_2,
-\]
-where $\theta_{N}$ stacks all the coefficients.
-
-\medskip
-\noindent\textbf{Step 2: Population risk and curvature penalty.}
-Define the population risk:
-\[
-  R(V)\;:=\;\mathbb{E}_{\rho}\!\bigl[H\!\bigl(X,\nabla V(X)\bigr)^{2}\bigr],
-  \qquad
-  C(V)\;:=\;\bigl\|D^{2}V(\bar x)-P\bigr\|^{2}_{F},
-\]
-and the regularised functional:
-\[
-  \mathcal J_{\lambda,\gamma}(V) := R(V)+\gamma C(V)+\lambda\|V\|_{\mathcal
↪  H_{k}}^{2}.
-\]
-Since $a(x)\succ 0$, the Hamiltonian $H$ is strictly convex in its second argument,
↪  ensuring that $V$ is the unique viscosity solution and also the unique minimiser of
↪  $\mathcal J_{\lambda,\gamma}$ (as $C(V)=0$ and the regularisers vanish at $V$).
-
-\medskip
-\noindent\textbf{Step 3: Empirical risk decomposition.}
-Define the empirical risk:
-\[
-  \widehat R_{N}(V)
-  :=\frac1N\sum_{p=1}^{N}
-     H\!\bigl(x_{p},\nabla V(x_{p})\bigr)^{2}.
-\]
-Then $V_{N}$ minimises the regularised empirical objective:
-\[
-  \widehat{\mathcal J}_{N}(V)
-    :=\widehat R_{N}(V)+\gamma_{N}C(V)+\lambda_{N}\|V\|_{\mathcal H_{k}}^{2}.
-\]
-By standard excess risk decomposition:
-\begin{align*}
-  R(V_{N})-R(V)
-  &= \bigl[R(V_{N})-\widehat R_{N}(V_{N})\bigr]
-   +\bigl[\widehat R_{N}(V_{N})-\widehat R_{N}(V)\bigr]
-   +\bigl[\widehat R_{N}(V)-R(V)\bigr].
```

```
-\end{align*}
-The middle term is $\le -\gamma_{N}C(V_{N}) - \lambda_{N}(\|V_{N}\|_{\mathcal
↪  H_{k}}^{2} - \|V\|_{\mathcal H_{k}}^{2})$ by optimality of $V_{N}$.
-Hence,
-\begin{equation}\label{eq:excess-risk-revised}
-  R(V_{N})-R(V) + \gamma_{N}C(V_{N})
-  \;\le\;
-  2\,\sup_{V\in\mathcal B_{N}}
-  \bigl|R(V)-\widehat R_{N}(V)\bigr|,
-\end{equation}
-
-\iffalse
-where we now define the constraint set:
-\[
-  \mathcal B_{N}
-  := \bigl\{ V\in\mathcal H_{k} :
-        \|V\|_{\mathcal H_{k}} \le C + \lambda_{N}^{-1/2} \bigr\}.
-\]
-\fi
-
-Define the comparison ball
-\[
-  \mathcal{B}_N := \left\{ f \in \mathcal{H}_k : \|f\|_{\mathcal{H}_k} \le
↪  \|V\|_{\mathcal{H}_k} + \lambda_N^{-1/2} \right\},
-\]
-\textcolor{red}{where $V$ denotes the true quasi-potential (as in the Theorem
↪  statement), so that $V_N \in \mathcal{B}_N$ with high probability.}
-
-
-\medskip
-\noindent\textbf{Step 4: Uniform deviation bound.}
-Since $k$ is bounded and smooth on compact $\Omega$, and $H(x,\cdot)$ is Lipschitz in
↪  $\nabla V(x)$, the Lipschitz constant over $\mathcal B_{N}$ is
↪  $O(\lambda_{N}^{-1/2})$. Talagrand-Bousquet inequality gives:
-\[
-  \sup_{V\in\mathcal B_{N}}
-  \bigl|R(V)-\widehat R_{N}(V)\bigr|
-  \le
-  C\left( \sqrt{\frac{\log(1/\delta)}{N}}\,\lambda_{N}^{-1/2}
-         + \frac{\log(1/\delta)}{N} \right)
-\]
-with high probability. Therefore,
-\[
-  \sup_{V\in\mathcal B_{N}} \bigl|R(V)-\widehat R_{N}(V)\bigr| =
↪  O_{p}\!\left(\sqrt{\frac{1}{N\lambda_{N}}}\right).
-\]
-Here, the notation $\mathcal{O}_p(\epsilon_N)$ means that the random variable is
↪  bounded in probability at rate $\epsilon_N$, i.e., for all $\delta > 0$ there
↪  exists $M > 0$ such that
-$\mathbb{P}(|Z_N| > M\epsilon_N) < \delta$ for all sufficiently large $N$.
-
-\medskip
-\noindent\textbf{Step 5: Convergence in risk and curvature.}
-The assumption $N\lambda_{N}\to\infty$ implies that the RHS of
↪  \eqref{eq:excess-risk-revised} vanishes in probability. Since
↪  $\gamma_{N}/\lambda_{N}=O(1)$ and $\gamma_{N}\to 0$, it follows that:
-\[
-R(V_N) - R(V) \to 0, \qquad C(V_N) \to 0 \quad \text{in probability}.
-\]
-
```

```
-\medskip
-\noindent\textbf{Step 6: Convergence in \boldmath$L^2$ and gradient.}
-
-The coercivity of $a(x)$ yields:
-\[
-  \tfrac12\lambda_{\min}(a)\,
-  \|\nabla V_{N}-\nabla V\|^{2}_{L^{2}(\rho)}
-  \;\le\; R(V_{N})-R(V),
-\]
-hence $\nabla V_{N} \to \nabla V$ in $L^2(\rho)$, and $V_N \to V$ in $L^2(\rho)$ by
↪  Poincaré-Friedrichs on bounded $\Omega$.
-
-\medskip
-\noindent\textbf{Step 7: Hessian consistency.}
-$C(V_N) \to 0$ implies
-\[
-D^2 V_N(\bar x) \to P \quad \text{in Frobenius norm, and hence in probability}.
-\]
-
-\medskip
-\noindent\textbf{Step 8: Uniform convergence (optional).}
-If $V\in C^m(\Omega)$ and $k\in C^m$, the $\mathcal \mathcal{H}_k$ norm controls $m$
↪  derivatives. By Sobolev embedding or Arzelà-Ascoli, convergence is uniform on
↪  compact subsets:
-\[
-V_N \to V \quad \text{in } C^{m-1}(K) \quad \text{for all compact } K \Subset  \Omega,
-\]
-where $K \Subset \Omega$ implies that $K$ is a compact subset whose closure lies
↪  strictly inside $\Omega$. This ensures that the convergence statements avoid
↪  potential boundary irregularities, which is standard in the analysis of convergence
↪  in function spaces.
-
-\vspace{-1.2\baselineskip}
+\begin{proof}[Proof of Theorem~\ref{thm:consistency-curvature}]
+\textbf{Step 1: Finite representations and their scope.}
+Kernel section expansions provide the numerical parametrizations. In an optimization
↪  over a scalar RKHS with bounded derivative evaluation functionals, a generalized
↪  representer theorem uses kernel derivatives associated with those functionals
↪  \cite{Chen2021}. A quadratic form in RKHS eigenfunctions is not automatically the
↪  same scalar RKHS minimizer. The convergence argument below uses the regularity and
↪  compactness stated in the theorem, rather than making that identification. In
↪  particular, bounded second-derivative evaluation functionals require more kernel
↪  regularity than merely assuming that the kernel is $C^2$.
+
+\medskip\noindent\textbf{Step 2: Population risk and curvature.}
+For any candidate $u$, define
+\[
+ R(u)=\mathbb E_\rho[H(X,\nabla u(X))^2],\qquad
+ C(u)=\|D^2u(\bar x)-P\|_F^2.
+\]
+The regularized population objective is $R(u)+\gamma C(u)+\lambda\mathcal R(u)$. Zero
↪  residual does not imply uniqueness, and a regularizer need not vanish at the true
↪  solution. Uniqueness of the viscosity limit will instead follow from the assumed
↪  boundary-value comparison principle.
+
+\medskip\noindent\textbf{Step 3: What empirical optimality gives.}
+Let $\widehat R_N(u)=N^{-1}\sum_p H(x_p,\nabla u(x_p))^2$. If $u_N$ minimizes the
↪  regularized empirical objective up to tolerance $\delta_N\geq0$, comparison with
↪  any admissible $w_N$ gives
```

```
+\begin{equation}\label{eq:excess-risk}
+\begin{aligned}
+\widehat R_N(u_N)+\gamma_N C(u_N)+\lambda_N\mathcal R(u_N)
+\leq{}&\widehat R_N(w_N)+\gamma_N C(w_N)\\
+&+\lambda_N\mathcal R(w_N)+\delta_N.
+\end{aligned}
+\end{equation}
+All comparator regularization and optimization terms must be retained. This inequality
↪  explains the need for approximation and optimization control; it does not give a
↪  gradient-error bound. The comparison with kernel-ridge excess-risk analysis
↪  \cite{CaponnettoDeVito2007,SteinwartChristmann2008} is methodological, not a
↪  transfer of its rates to the nonlinear HJ loss.
+
+\medskip\noindent\textbf{Step 4: From samples to a uniform residual.}
+For each $x\in\overline D$, choose a nearest collocation point. The common modulus
↪  yields
+\begin{equation}\label{eq:uniform-residual}
+ \|r_N\|_{L^\infty(D)}
+ \leq \max_{x_p\in X_N}|r_N(x_p)|+\omega(h_N)\longrightarrow0.
+\end{equation}
+This is a stronger requirement than $\widehat R_N(V_N)\to0$. For example,
+$\max_p|r_N(x_p)|\leq\sqrt{N\widehat R_N(V_N)}$ gives the sufficient, but not
↪  necessary, condition $N\widehat R_N(V_N)\to0$ when combined with uniform residual
↪  regularity. A population-risk route would need a uniform generalization estimate
↪  and sufficient control of residual spikes; no such rate is asserted here.
+
+\medskip\noindent\textbf{Step 5: Compactness.}
+Uniform boundedness and equicontinuity give, by Arzel\`a--Ascoli, a uniformly
↪  convergent subsequence $V_{N_j}\to u$ on $\overline D$. The boundary assumption
↪  implies $u=g$ on $\partial D$. In particular, compactness is an independent
↪  hypothesis; kernel smoothness alone does not provide a uniform bound on fitted
↪  derivatives as regularization vanishes.
+
+\medskip\noindent\textbf{Step 6: Viscosity identification.}
+If a $C^1$ function $\phi$ touches $u$ from above at an interior point $x$, a small
↪  strictifying perturbation and uniform convergence produce local maxima of
↪  $V_{N_j}-\phi$ at $x_j\to x$. There,
+\[
+ \nabla V_{N_j}(x_j)=\nabla\phi(x_j),\qquad
+ H(x_j,\nabla\phi(x_j))=r_{N_j}(x_j)\to0.
+\]
+Continuity of $H$ gives the subsolution inequality. The same argument with local
↪  minima gives the supersolution inequality. Thus $u$ is a viscosity solution with
↪  data $g$. Comparison identifies $u=V$. Every subsequential limit has this value, so
↪  the entire sequence converges uniformly and therefore in $L^2(\rho)$.
+
+\medskip\noindent\textbf{Step 7: Hessian consistency.}
+The additional curvature hypothesis gives the stated Hessian convergence directly. For
↪  a penalized fit it is sufficient that the attained nonnegative objective in
↪  \eqref{eq:empirical-loss} is $o(\gamma_N)$, because
+\[
+ C(V_N)\leq\mathcal L_{N,\lambda_N,\gamma_N}(V_N)/\gamma_N.
+\]
+Merely taking $\gamma_N\to0$ does not imply this conclusion. Exact curvature
↪  constraints are an alternative.
+
+\medskip\noindent\textbf{Step 8: The additional gradient conclusion.}
```

```
+Under the stated gradient compactness assumption, any subsequence has a further
↪  subsequence whose gradients converge uniformly on $K$ to some $q$. Integration
↪  against compactly supported test functions, together with uniform convergence of
↪  $V_N$, identifies $q$ as the distributional gradient of $V$, hence as $\nabla V$
↪  when $V\in C^1$. Uniqueness of this limit gives convergence on every compact
↪  subset. No gradient convergence is asserted across a nonsmooth quasi-potential
↪  without further analysis.
 \end{proof}

-\vspace{0.5em}
-\noindent
-
-Theorem~\ref{thm:consistency-curvature} establishes that the learned quasi-potential
↪  $V_N$ converges to the true solution $V$ of the Hamilton--Jacobi equation in both
↪  value and gradient. In particular, the penalty term $\|D^2 V_N(\bar{x}) - P\|_F^2$
↪  enforces curvature consistency at the stable equilibrium $\bar{x}$. To clarify the
↪  necessity and interpretation of this Riccati-based curvature constraint, we include
↪  the following corollary, which shows that any sufficiently regular solution of the
↪  Hamilton--Jacobi equation must satisfy this Riccati condition locally at $\bar{x}$.
-
-\begin{corollary}[Local Consistency with the Hamilton-Jacobi
↪  Equation]\label{cor:localHJ}
+\paragraph{Interpretation of the assumptions.}
+For a quasi-potential, the boundary or admissibility problem must select the action
↪  value in \eqref{quasip}. A truncated, punctured basin with compatible action data
↪  is one possible setting, provided its comparison principle is established. The
↪  local experimental fits do not establish those conditions and are therefore
↪  assessed through numerical diagnostics rather than claimed as instances of an
↪  unconditional consistency theorem. Likewise, the monotone-scheme theorem of Barles
↪  and Souganidis \cite{Barles1991} cannot be applied directly: monotonicity of the
↪  present nonlinear kernel least-squares scheme has not been proved. A fixed
↪  quadratic Koopman class need not be dense in the required derivative topology even
↪  when the scalar kernel is universal. These distinctions preserve the role of
↪  Riccati matching without equating local curvature with global solution selection.
+
+The following corollary records the local Riccati condition for a smooth solution with
↪  a nondegenerate minimum. It is separate from the global viscosity-selection
↪  question.
+
+\begin{corollary}[Local Consistency with the Hamilton--Jacobi
↪  Equation]\label{cor:localHJ}
 Let $V \in C^{2}(\Omega)$ solve
 \[
 H(x,\nabla V(x))
@@ -507,14 +409,14 @@
 +\tfrac{1}{2}\nabla V(x)^\top a(x)\,\nabla V(x)=0,
 \qquad V(\bar{x})=0,
 \]
-with $\bar{x}$ a globally asymptotically stable equilibrium of $f$.
+with $f=b$, $b\in C^1$, $a(\bar x)\succ 0$, $Db(\bar x)$ Hurwitz, $\nabla V(\bar x)=0$,
↪  and $D^2V(\bar x)\succ 0$. The positive-Hessian condition excludes the constant HJ
↪  solution.
 Then the Hessian of $V$ at $\bar{x}$ equals the Riccati matrix $P$:
 \[
 D^{2}V(\bar{x})=P,
 \]
 where $P$ is the unique positive-definite solution to the algebraic Riccati equation
 \[
-b'(\bar{x})^\top P + P\,b'(\bar{x}) = P\,a(\bar{x})\,P.
```

```
+b'(\bar{x})^\top P + P\,b'(\bar{x}) + P\,a(\bar{x})\,P=0.
 \]
 \end{corollary}

@@ -531,7 +433,7 @@
 a(x)=a(\bar{x})+o(1).
 \]
 Thus the gradient satisfies $\nabla V(x)=Z(x-\bar{x})+o(\|x-\bar{x}\|)$.
-Insert these expansions into the Hamilton-Jacobi equation:
+Insert these expansions into the Hamilton--Jacobi equation:
 \[
 H(x,\nabla V(x))=
 f(x)^\top\nabla V(x)
@@ -549,633 +451,293 @@
 b'(\bar{x})^\top Z + Z b'(\bar{x}) + Z a(\bar{x})Z =0.
 \]
 By standard results on algebraic Riccati equations (positive-definite $a(\bar{x})$ and
 ↪  stability of $b'(\bar{x})$), there is a unique positive-definite solution $P$ of
 ↪  this equation.
-Hence $Z=P$ and therefore $D^{2}V(\bar{x})=P$.
+Since $Z\succ0$, multiplying by $Z^{-1}$ gives the uniquely solvable Lyapunov equation
↪  for $Z^{-1}$; hence $Z=P$ and $D^{2}V(\bar{x})=P$.
 \end{proof}



-\section{Examples}
-\label{examples}
-
-\subsection{Example 1: The two-Dimensional Maier--Stein System}\label{sec:example1}
-
-We consider the Maier--Stein system whose deterministic part is
-\[
-\dot x = F(x) =
-\begin{pmatrix}
-x_1 - x_1^3 - x_1 x_2^2 \\[6pt]
--(1 + x_1^2) x_2
-\end{pmatrix},
-\]
-which features a stable fixed point at \( \bar{x} = (-1,0) \). Throughout this
↪  example, we assume the diffusion matrix is the identity, i.e., $a(x) \equiv I$.
↪  Note that the corresponding deterministic system is a gradient system with the
↪  potential
-\[
-U(x) = \frac{1}{4} (x_1^2 - 1)^2 + \frac{1}{2} (x_1^2 + 1)x_2^2.
-\]
-\textcolor{red}{The associated quasi-potential in this case is given by (here and
↪  throughout the examples, $V_{\rm true}$ denotes the analytically known
↪  quasi-potential, i.e., the true solution $V$ from Theorem
↪  \ref{thm:consistency-curvature})}
-\[
-V_{\rm true}(x) = 2U(x) = \frac{1}{2} (x_1^2 - 1)^2 + (x_1^2 + 1)x_2^2.
-\]
-Linearization at the fixed point gives the Jacobian
-\[
-J = D F(\bar{x}), \quad J w_i = \lambda_i w_i,
-\]
-where \( (\lambda_i, w_i) \) are the eigenpairs of \( J \). We solve the Lyapunov
↪  equation
-\[
```

```
-J Z + Z J^\top = -I, \quad \Rightarrow \quad P_{\rm loc} = Z^{-1},
-\]
-yielding \( P_{\rm loc} = \nabla^2 V(\bar{x}) \), the Hessian of the quasi-potential
↪  at the saddle.
-
-We seek eigenfunctions \( \varphi_i(x) \) satisfying the Koopman PDE:
-\[
-\nabla\varphi_i(x) \cdot F(x) = \lambda_i \varphi_i(x), \quad
-\varphi_i(\bar{x}) = 0, \quad \nabla\varphi_i(\bar{x}) = e_i.
-\]
-\textcolor{red}{Each \( \varphi_i \) is represented in an RKHS:}
-\[
-\varphi_i(x)
-= \sum_{p=1}^N \alpha^{(i)}_p k_\beta(x,x_p) + (x - \bar{x}) \cdot w_i,
-\quad
-k_\beta(x,t) = \sum_{j=1}^L \beta_j k_j(x,t).
-\]
-On a collocation set \( \{x_p\}_{p=1}^N \), we impose the constraints:
-\begin{align*}
-&\varphi_i(\bar{x}) = 0, \\
-&\nabla\varphi_i(\bar{x}) \cdot e_j = \delta_{ij}, \quad j=1,2, \\
-&\nabla\varphi_i(x_p) \cdot F(x_p) = \lambda_i (x_p - \bar{x}) \cdot w_i, \quad p = 1,
↪  \dots, N.
-\end{align*}
-Solving the regularized linear system yields the coefficients \( \alpha^{(i)} \).
-
-Define \( \Phi(x) = (\varphi_1(x), \varphi_2(x))^\top \), and construct the
↪  quasi-potential:
-\[
-V_{\rm mid}(x) = \tfrac12\,\Phi(x)^T P_{\rm mid} \Phi(x)
-\]
-with
-\[
-P_{\rm mid} = W\,\mathrm{diag}(-2\lambda_i)\,W^{-1}.
-\]
-
-\textcolor{red}{To determine \( \beta \), we minimize the composite
↪  loss~\eqref{eq:composite_loss} (the same $L_{\text{HJ}}$, $L_{\text{curv}}$ terms
↪  defined in Section~\ref{methsec}):}
-\[
-L(\beta) = L_{\text{HJ}} + \gamma L_{\text{curv}} + \lambda_1 \sum_{j=1}^L |\beta_j| +
↪  \eta \sum_{i=1}^2 \|\varphi_i\|_{\mathcal{H}_k}^2,
-\]
-where
-\[
-L_{\text{HJ}} = \sum_{p=1}^N \left[ F(x_p) \cdot \nabla V_{\text{mid}}(x_p)
-+ \tfrac12 \|\nabla V_{\rm mid}(x_p)\|^2 \right]^2,
-\]
-\[
-L_{\text{curv}} = \|\nabla^2 V_{\text{mid}}(\bar{x}) - P_{\text{loc}}\|_F^2,
-\]
-\[
-\|\varphi_i\|_{\mathcal{H}_k}^2 = \alpha^{(i)\,T} G_\beta \alpha^{(i)}, \quad
↪  (G_\beta)_{pq} = k_\beta(x_p, x_q).
-\]
-
-Let $\beta^*$ denote the optimal (possibly sparse) coefficient vector such that the
↪  true solution $V$ can be well approximated within the model class spanned by the
↪  fixed dictionary. After obtaining \( \beta^* \), we:
-\begin{itemize}
```

```
-  \item Threshold small coefficients \( \beta_j < 1/L \) to zero and renormalize.
-  \item Recompute \( \varphi_i \).
-  \item Rescale \( V_{\rm mid} \) via
-    \[
-    s = \frac{\langle V_{\rm true}, V_{\rm mid} \rangle}{\|V_{\rm mid}\|^2}, \quad
-    \text{RMSE} = \sqrt{\frac{1}{n^2} \sum (s V_{\rm mid} - V_{\rm true})^2}.
-    \]
-\end{itemize}
-By sweeping parameters \( \lambda_1 \), \( \gamma \), and \( \eta \), we observe
↪  trade-offs between sparsity, curvature matching, and PDE residual minimization.
-
-
-
-\textcolor{red}{The choice of kernel dictionary (two Gaussian RBFs and five polynomial
↪  kernels) is discussed in Section~3; see the ``Kernel dictionary'' paragraph
↪  therein.}
-
-\paragraph{Numerical Setup.}
-We linearize around \( \bar x = (-1, 0) \). Since the eigenvalues of \( J \) are both
↪  \( -2 \), the fixed point is stable, and the Lyapunov equation is solvable. We
↪  construct an \( q \times q \) uniform grid around \( \bar{x} \) (e.g., \( q = 10
↪  \)) and define \( L = 7 \) kernel functions:
-\begin{itemize}
-  \item Two Gaussian RBFs: \( \exp(-\|x - y\|^2 / (2\sigma^2)) \) with \( \sigma =
↪  0.5, 1.5 \),
-  \item Five polynomial kernels: \( (1 + x \cdot y)^r \), for \( r = 2, 3, 4, 5, 6 \).
-\end{itemize}
-
-
-
-\paragraph{Collocation point distribution.}
-We construct an $q \times q$ uniform grid around $\bar{x}$ (e.g., $q=10$) rather than
↪  aligning the grid with the eigenvectors of the Jacobian $J = \nabla b(\bar{x})$.
↪  This choice avoids two potential issues: (i) axis-aligned grids can lead to
↪  ill-conditioned kernel Gram matrices when the dynamics have vastly different time
↪  scales along different eigendirections, and (ii) strict alignment may undersample
↪  regions where the quasi-potential has important features that are not aligned with
↪  the linearized eigendirections, particularly away from the equilibrium where
↪  nonlinear effects dominate. A uniform or randomly distributed collocation set
↪  ensures more robust geometric coverage of the entire domain.
-
-
-We define the convex combination:
-\[
-k_\beta(x, y) = \sum_{j=1}^L \beta_j^2 k_j(x, y).
-\]
-\textcolor{red}{The kernel weights satisfy $\sum_j \beta_j^2 = 1$.}
-For each \( i = 1, 2 \), we enforce:
-\begin{enumerate}
-  \item \( \varphi_i(\bar x) = 0 \),
-  \item \( \nabla \varphi_i(\bar x) = e_i \),
-  \item PDE residual: \( \nabla \varphi_i(x_p) \cdot F(x_p) + \lambda_i (x_p - \bar x)
↪  \cdot w_i = 0 \).
-\end{enumerate}
-
-We solve the regularized system to obtain \( \beta \), then construct:
-\[
-\Phi = [\varphi_1, \varphi_2]^\top, \quad
-P_{\mathrm{mid}} = W\,\mathrm{diag}(-2\lambda_i)W^{-1}, \quad
-V_{\mathrm{mid}}(x) = \tfrac12 \Phi(x)^\top P_{\mathrm{mid}} \Phi(x).
```

```
-\]
-
-To promote sparsity, we minimize:
-\[
-\|\text{HJ residual}\|_2 + \gamma_{\text{curv}} \|\nabla^2 V_{\text{mid}}(\bar{x}) -
↪  P_{\mathrm{loc}}\|_F + \lambda_1 \|\beta\|_1,
-\]
-Sweeping \( \lambda_1 \in \{0, 0.01, 0.1, 1, 10, 50, 100\} \), we observe:
-\begin{itemize}
-  \item \( \lambda_1 \le 10 \): All 7 kernels active, RMSE \( \approx 30.4 \).
-  \item \textcolor{red}{\( \lambda_1 = 50 \): 3 kernels remain (small RBF, polynomial
↪  \( r = 2 \), polynomial \( r = 4 \)), weights \( [0.165, 0.271, 0.565] \), RMSE \(
↪  \approx 30.6 \).}
-  \item \textcolor{red}{\( \lambda_1 = 100 \): 3 kernels (wide RBF, polynomial \( r =
↪  2 \), polynomial \( r = 3 \)), RMSE \( \approx 58.1 \).}
-\end{itemize}
-
-\textcolor{red}{The optimal trade-off in the sense of sparsity--accuracy balance
↪  occurs at $\lambda_1 = 50$. As shown in Figure~\ref{fig:rmse_vs_lambda1}, the
↪  numerically smallest RMSE is attained for $\lambda_1 \leq 10$, but these small
↪  values keep all seven kernels active (Figure~\ref{fig:pruned_kernels_vs_lambda1}),
↪  providing no model reduction. In contrast, $\lambda_1 = 50$ removes four kernels
↪  while maintaining nearly the same RMSE ($\approx 30.6$ vs.\ $\approx 30.4$) and
↪  producing a reconstructed potential visually indistinguishable from the true
↪  potential (Figure~\ref{fig:all_V_mid_surfaces}). Thus, $\lambda_1 = 50$ lies near
↪  the ``knee'' of the sparsity--accuracy curve: it achieves the best balance between
↪  parsimony (reducing model complexity by $>50\%$) and approximation quality. Larger
↪  values of $\lambda_1$ lead to over-sparsification and visible degradation in the
↪  reconstructed potential, while smaller values provide only marginal improvements in
↪  RMSE (less than $1\%$) at the cost of retaining unnecessary kernels.}
-
-\textcolor{red}{Figures~\ref{fig:pruned_kernels_vs_lambda1}--\ref{fig:all_V_mid_surfa
↪  ces} visualize the effect of the sparsity parameter $\lambda_1$ on kernel selection
↪  and on the resulting quasi-potential approximation.
↪  Figure~\ref{fig:pruned_kernels_vs_lambda1} shows the number of active kernels as
↪  $\lambda_1$ varies; Figure~\ref{fig:rmse_vs_lambda1} displays the corresponding
↪  reconstruction RMSE, which remains nearly flat over a wide range of $\lambda_1$
↪  even as many kernels are pruned; and Figure~\ref{fig:all_V_mid_surfaces} presents
↪  reconstructed quasi-potential surfaces $V_{\mathrm{mid}}$ for each $\lambda_1$
↪  alongside the true potential. Together, these figures show that the hybrid method
↪  achieves accurate reconstruction while automatically pruning the kernel
↪  dictionary.}
-
-
-\begin{figure}[ht]
-    \centering
-    \includegraphics[width=0.7\linewidth]{Figure_1.png}
-    \caption{Number of active kernels as a function of the sparsity parameter
↪  $\lambda_1$.}
-    \label{fig:pruned_kernels_vs_lambda1}
+\paragraph{Computational complexity, memory, and dimensional
↪  scope.}\label{sec:complexity-full}
```

```
+Let $N$ be the collocation count, $M$ the number of centers ($M=N$ in a full section
↪  expansion), $n$ the state dimension, $L$ the dictionary size, and $T$ the number of
↪  outer kernel evaluations. For common radial kernels, constructing values and first
↪  derivatives costs $O(LNMn)$; retaining all derivative arrays uses $O(LNMn)$
↪  storage. A single dense $N\times N$ Gram matrix requires $8N^2$ bytes in float64.
↪  The often-quoted $O(N^2n)$ is an assembly cost, not the cost of solving a dense
↪  system: factorization requires $O(N^3)$ time and $O(N^2)$ memory. Initial
↪  computation of the Riccati target through a dense Lyapunov solve costs $O(n^3)$.
+
+For $n$ independent scalar eigenfunction fits with $M$ centers, dense least-squares
↪  work can be $O(n(NM^2+M^3))$ per kernel choice, in addition to assembly.
↪  Factorizations can sometimes be reused, but cannot generally be shared across
↪  distinct eigenvalues or updated kernels. Real block formulations for complex
↪  eigenpairs may require coupled solves. A generic fully coupled system has $Nn$ rows
↪  and $Mn$ columns: storing its design and normal matrices costs $O(NMn^2+M^2n^2)$,
↪  and forming/factoring normal equations can cost $O(NM^2n^3+M^3n^3)$. Exploiting
↪  spectral or block structure reduces this upper bound. The implementation in
↪  Appendix~\ref{app:additional} uses the coupled real form.
+
+A direct scalar $M$-coefficient HJ fit has nonlinear least-squares steps costing
↪  approximately $O(NM^2+M^3)$, plus gradient/residual assembly. Dense diffusion
↪  contributes $O(Nn^2)$ matrix-vector work, excluding coefficient evaluation. For
↪  structured radial derivatives, evaluating a trained spectral potential and gradient
↪  at one point costs approximately $O(LMn^2+n^2)$, versus $O(LMn+n^2)$ for a scalar
↪  section expansion. Dense Taylor-Hessian contractions in the exact-curvature direct
↪  implementation of Appendix~\ref{app:additional} add $O(Mn^2)$ work per point. The
↪  outer selection loop repeats inner work $T$ times. Derivative workspaces, optimizer
↪  arrays, and validation evaluation must therefore be included in practical budgets.
+
+Kernel sparsification reduces $L$ but does not by itself reduce $N$ or $M$. Nor does
↪  algebraic dependence on dimension in an individual operation remove the
↪  sample-coverage burden: a tensor grid with $q$ points per coordinate has $N=q^n$.
↪  The examples in Section~\ref{examples} cover dimensions two and three.
↪  Appendix~\ref{app:additional} studies a structured eight-dimensional system and
↪  reports timing and storage measurements; these tests do not establish
↪  dimension-independent scalability. Sparse centers, structured linear algebra, and
↪  reduced-order models are possible future improvements, rather than established
↪  performance guarantees of the current method.
+
+\subsection{Direct scalar implementation and stable finite
↪  features}\label{sec:direct-implementation}
+The HJ equation can be fitted directly in a scalar kernel space while retaining the
↪  same local Riccati information. This option separates potential approximation from
↪  the additional transport restriction imposed by a Koopman-coordinate model. Let
↪  $y=x-\bar x$. For polynomial kernels of degree $r$, use the finite-feature
↪  parametrization
+\begin{equation}\label{eq:direct-polynomial}
+ V_r(x)=\tfrac12y^\top Py+\sum_{3\leq|\nu|\leq r}c_\nu y^\nu.
+\end{equation}
+Here $\nu$ is a multi-index and $y^\nu=\prod_i y_i^{\nu_i}$. The space of polynomials
↪  of degree at most $r$ is the RKHS of the inhomogeneous polynomial kernel $(1+x\cdot
↪  z)^r$, with a degree-dependent inner product. Changing to centered monomials
↪  preserves this finite-dimensional function space. The coefficient penalty used in
↪  the implementation is explicitly Euclidean in these centered coordinates; it is not
↪  identified with the canonical polynomial-kernel norm.
+
+Restricting the correction to degrees at least three makes the value, gradient, and
↪  Hessian constraints exact. The coefficients minimize
+\begin{equation}\label{eq:direct-objective}
```

```
+ \mathcal J(c)=\frac1N\sum_{p=1}^N
+ H(x_p,\nabla V_r(x_p))^2+\eta\|c\|_2^2.
+\end{equation}
+Writing $g_p=\nabla V_r(x_p)$ and $D_{p\nu}=\nabla y_p^\nu$, the residual Jacobian is
+\[
+ \frac{\partial H(x_p,g_p)}{\partial c_\nu}
+ =\bigl(b(x_p)+a(x_p)g_p\bigr)^\top D_{p\nu}.
+\]
+This analytic derivative avoids differencing a nearly singular kernel linear solve.
↪  Columns are scaled by the training RMS of their derivatives for numerical
↪  conditioning; the transformation is undone when storing coefficients and evaluating
↪  the regularizer. Such scaling changes the optimization coordinates, not the fitted
↪  function or the objective.
+
+In two dimensions, the number of free coefficients in \eqref{eq:direct-polynomial} is
↪  $\binom{r+2}{2}-6$. The degree-two candidate has no correction and equals the
↪  Riccati baseline. Candidate degree selection uses independent HJ residuals, with
↪  the precise tolerance and continuation schedule specified in
↪  Section~\ref{examples}. Analytic potential values are unnecessary for these
↪  choices.
+
+\paragraph{An anchored quadratic kernel expansion.}
+For the three-dimensional degree-two kernel, subtracting its value at the equilibrium
↪  gives a nine-dimensional function space. At $\bar x=0$, write
+\[
+ q(x)=(x_1,x_2,x_3,x_1^2,x_2^2,x_3^2,x_1x_2,x_1x_3,x_2x_3)^\top,
+ \qquad V_h(x)=q(x)^\top c.
+\]
+Expanding $(1+x\cdot z_j)^2-1$ gives a matrix $B\in\mathbb R^{9\times N}$ such that
↪  $c=B\alpha$. Its first three rows are $2z_{ji}$, the next three are $z_{ji}^2$, and
↪  the remaining rows are the cross products $2z_{ji}z_{jk}$. When $B$ has full row
↪  rank, every anchored quadratic is represented and the minimum Euclidean-norm kernel
↪  coefficient vector is $\alpha=B^\dagger c$. Therefore optimizing
+\[
+ \frac1N\sum_p H(x_p,\nabla(q^\top c))^2
+ +\gamma\Bigl\|\sum_\nu c_\nu D^2q_\nu-P\Bigr\|_F^2
+ +\eta\|B^\dagger c\|_2^2
+\]
+is exactly a minimum-norm coefficient refit in the anchored kernel class, with the
↪  mixture weight fixed at one. It uses nine identifiable variables instead of $N$
↪  redundant variables and enforces the additive normalization algebraically. Singular
↪  values of $B$ determine its numerical rank; all sample sizes used in
↪  Section~\ref{examples} have rank nine.
+
+These direct parametrizations and the spectral construction share HJ residual fitting
↪  and local curvature information, but impose different global restrictions. Their
↪  controlled comparison in Appendix~\ref{app:additional} evaluates that distinction.
↪  Finite-feature compression is particularly effective when a selected polynomial
↪  kernel has small rank; it does not remove the dense-kernel cost for a general RBF
↪  representation.
+
+\section{Examples}\label{examples}
+\paragraph{Numerical protocol.}\label{sec:numerical-scope}
```

```
+We evaluate three SDEs with analytic reference potentials. The first two tests use the
↪  direct scalar formulation in Section~\ref{sec:direct-implementation}, with
↪  polynomial kernels of degrees $r=2,\ldots,6$. The third uses an anchored degree-two
↪  kernel expansion. These experiments measure approximation accuracy, local
↪  constraints, model selection, and sample-count sensitivity.
↪  Appendix~\ref{app:additional} separately compares direct and spectral
↪  representations on common data, including an eight-dimensional system.
+
+The analytic potential is used only for evaluation. Training uses the drift,
↪  diffusion, and the Riccati Hessian computed from the equilibrium linearization. No
↪  scalar amplitude is fitted to the reference potential. All computations use float64
↪  arithmetic. For the two-dimensional examples, the training seeds are $42,43,44$,
↪  each with $N=200$ uniform points. Independent validation sets contain $2048$
↪  points, with seeds $10000+s$ for training seed $s$. A common set of $10000$
↪  independent test points, generated with seed $20260923$, evaluates the three fits.
↪  Sharing this test set separates variation in the fitted model from variation in the
↪  numerical integration of its error.
+
+We use four fixed continuation radii $t\in\{1/8,1/4,1/2,1\}$. At radius $t$, the
↪  collocation locations are $\bar x+t(x_p-\bar x)$, and the previous solution
↪  initializes the next fit. Thus each candidate uses $4N$ distinct collocation
↪  locations before validation. At the first stage, the correction coefficients are
↪  zero, giving the Riccati warm start. This continuation connects the local
↪  nontrivial branch to the larger training domain. The final objective uses mean
↪  squared HJ residual plus $10^{-8}\|c\|_2^2$. Analytic Jacobians are supplied to
↪  nonlinear least squares, with tolerances $10^{-13}$ for function and step
↪  termination, $10^{-10}$ for gradient termination, and a maximum of $3000$ residual
↪  evaluations per stage.
+
+The selection rule is fixed before reference errors are evaluated: choose the smallest
↪  polynomial degree whose independent validation HJ RMS is at most $10^{-6}$ and
↪  whose four stages terminate successfully. If none meets this condition, choose the
↪  smallest validation HJ RMS. This is a discrete complexity selection procedure; the
↪  amplitude penalty in \eqref{eq:composite_loss} describes the broader mixture
↪  formulation. The numerical tables identify the particular parametrization used in
↪  each experiment.
+
+\subsection{Example 1: The two-dimensional Maier--Stein system}\label{sec:example1}
+Consider the Maier--Stein model with coupling parameter equal to one:
+\begin{equation}\label{eq:ms-sde}
+\begin{aligned}
+ dx_1&=(x_1-x_1^3-x_1x_2^2)\,dt+\sqrt\varepsilon\,dW_1,\\
+ dx_2&=-(1+x_1^2)x_2\,dt+\sqrt\varepsilon\,dW_2.
+\end{aligned}
+\end{equation}
+The stable equilibrium of interest is $\bar x=(-1,0)$, with $J=-2I$ and $a=I$. Solving
↪  $JC+CJ^\top=-I$ gives $C=I/4$ and $P=C^{-1}=4I$. Consequently,
+\[
+ V_Q(x)=2\bigl((x_1+1)^2+x_2^2\bigr).
+\]
+The basin is $x_1<0$. The gradient structure gives the reference potential
+\begin{equation}\label{eq:ms-reference}
+ V_*(x)=\tfrac12(x_1^2-1)^2+(x_1^2+1)x_2^2.
+\end{equation}
+Direct differentiation yields $\nabla V_*=-2b$ and hence $H(x,\nabla V_*)=0$; the
↪  normalization at $(-1,0)$ fixes its constant. Values on the basin boundary are
↪  understood by continuous extension.
+
+\paragraph{Kernel reconstruction and selection.}
```

```
+The training and validation domain is $[-4,0]\times[-3,3]$. Polynomial kernels contain
↪  the quartic reference, but the degree is selected by the HJ validation criterion
↪  rather than by reference error. Writing $y=x-\bar x$, the fitted form is
+\[
+ V_r(x)=\tfrac12y^\top Py+\sum_{3\leq|\nu|\leq r}c_\nu y^\nu.
+\]
+For $r=2$ this reduces to the Riccati approximation. The cubic model supplies four
↪  free coefficients; degrees four, five, and six supply nine, fifteen, and
↪  twenty-two, respectively. Every model has the prescribed value, gradient, and
↪  Hessian at the equilibrium. These exact local constraints prevent the constant zero
↪  solution from being represented.
+
+All three seeds select $r=4$. For seed 42, the principal correction coefficients are
↪  approximately
+\[
+ c_{30}=-2,\qquad c_{12}=-2,\qquad c_{40}=\tfrac12,\qquad c_{22}=1,
+\]
+while the other degree-three and degree-four coefficients are close to zero. These
↪  coefficients are obtained by HJ fitting; expanding \eqref{eq:ms-reference}
↪  afterward confirms their interpretation. In this representable benchmark, the small
↪  residual reflects both the polynomial structure of the solution and the stable
↪  local constraints.
+
+\paragraph{Accuracy and spatial diagnostics.}
+On the uniform $10\times10$ evaluation grid over $[-4,0]\times[-3,3]$, the potential
↪  RMSEs for seeds $42,43,44$ are $1.13\times10^{-10}$, $1.55\times10^{-10}$, and
↪  $1.62\times10^{-10}$. The independent random tests give the accuracy and HJ
↪  diagnostics recorded in the supplied numerical tables and arrays. The Riccati
↪  approximation is a local quadratic, so its error increases substantially away from
↪  the equilibrium on this broad box. The quartic fit captures the nonlinear increase
↪  in potential without reference-based rescaling.
+
+Figures~\ref{fig:pruned_kernels_vs_lambda1} and~\ref{fig:rmse_vs_lambda1} display the
↪  degree comparison for both two-dimensional systems. The first plot contains the
↪  model-selection quantities; the second evaluates potential error only after the
↪  candidates have been fitted. Increasing polynomial degree beyond four is
↪  unnecessary for these two benchmarks and can increase conditioning and estimation
↪  sensitivity. Figure~\ref{fig:all_V_mid_surfaces} compares the Maier--Stein
↪  surfaces, and Figure~\ref{fig:residuals_example1} separates potential error from
↪  the HJ residual. Each residual panel has its own labeled scale.
+
+\begin{figure}[htbp]\centering
+\includegraphics[width=\linewidth]{figures/native/Figure_1.pdf}
+\caption{Independent validation HJ RMS versus polynomial kernel degree for three
↪  training seeds. The dashed line is the fixed selection tolerance $10^{-6}$. Both
↪  systems select the smallest acceptable degree,
↪  $r=4$.}\label{fig:pruned_kernels_vs_lambda1}
 \end{figure}
-
-\begin{figure}[ht]
-    \centering
-    \includegraphics[width=0.7\linewidth]{Figure_2.png}
-    \caption{Root-mean-square error (RMSE) of the quasi-potential approximation versus
↪  $\lambda_1$.}
-    \label{fig:rmse_vs_lambda1}
+\begin{figure}[htbp]\centering
+\includegraphics[width=\linewidth]{figures/native/Figure_2.pdf}
```

```
+\caption{Potential RMSE for the fitted polynomial candidates. The Maier--Stein grid is
↪  $10^2$ points on $[-4,0]\times[-3,3]$; the double-well grid is $101^2$ points on
↪  $[-1.5,1.5]^2$. The latter measures error against the polynomial reference
↪  extension, including extrapolation outside the training basin. Potential errors are
↪  not used for model selection.}\label{fig:rmse_vs_lambda1}
 \end{figure}
-
-\begin{figure}[ht]
-    \centering
-    % NOTE: Ensure sufficient vertical spacing between subfigures to eliminate
↪  overlap.
-    \includegraphics[width=1.0\linewidth]{Figure_3.png}
-    \caption{\textcolor{red}{Reconstructed quasi-potential surfaces $V_{\mathrm{mid}}$
↪  for each $\lambda_1$ alongside the true potential (bottom-right panel). The
↪  absolute pointwise error $|V_{\mathrm{mid}} - V_{\mathrm{true}}|$ for each
↪  $\lambda_1$ is shown in Figure~\ref{fig:residuals_example1}.}}
-    \label{fig:all_V_mid_surfaces}
+\begin{figure}[htbp]\centering
+\includegraphics[width=\linewidth]{figures/native/Figure_3.pdf}
+\caption{Maier--Stein potential on the left-basin closure: analytic reference, local
↪  Riccati quadratic, and the degree-four kernel fit with seed 42. Surfaces use an
↪  $81^2$ display grid; the fit uses 200 training points at each continuation
↪  radius.}\label{fig:all_V_mid_surfaces}
 \end{figure}
-
-% Figure showing spatial residuals for Example 1
-\begin{figure}[ht]
-    \centering
-    \includegraphics[width=1.0\linewidth]{Figure_4_residuals.png}
-    \caption{\textcolor{red}{Spatial residuals $|V_{\mathrm{mid}} -
↪  V_{\mathrm{true}}|$ for each value of $\lambda_1 \in \{0, 0.01, 0.1, 1, 10, 50,
↪  100\}$ in Example~1 (Maier--Stein, $\gamma=1$), shown on the domain
↪  $[-2.5,\,0.2]\times[-1.8,\,1.8]$. A shared colorscale facilitates direct comparison
↪  of error magnitude and spatial structure across panels. The RMSE of the Riccati
↪  (local quadratic) baseline is $5.39$; the Hybrid Kernel method achieves its best
↪  RMSE of $4.27$ at $\lambda_1=50$, a $21\%$ improvement, with residuals concentrated
↪  near the domain boundary rather than the equilibrium~$\bar{x}$ (white cross). For
↪  $\lambda_1=100$, over-sparsification removes too many kernel components, slightly
↪  degrading accuracy.}}
-    \label{fig:residuals_example1}
+\begin{figure}[htbp]\centering
+\includegraphics[width=\linewidth]{figures/native/Figure_4_residuals.pdf}
+\caption{Maier--Stein absolute potential error and absolute HJ residual on the $81^2$
↪  display grid. The top row is the Riccati baseline and the bottom row the
↪  degree-four fit. Color scales are stated separately because the errors differ by
↪  many orders of magnitude.}\label{fig:residuals_example1}
 \end{figure}
-\paragraph{Constraint satisfaction diagnostics.}
-To verify that the optimized surrogate satisfies the constraints, we report the
↪  following diagnostics for $\lambda_1 = 50$ (the optimal sparsity-accuracy
↪  trade-off):
-
-\begin{itemize}
-\item \textbf{Hamilton--Jacobi residual:} Mean $|\mathcal{L}_{\text{HJ}}| = 1.8 \times
↪  10^{-2}$ with maximum $8.3 \times 10^{-2}$ across the collocation grid.
-
-\item \textbf{Boundary condition:} $V_{\text{mid}}(\bar{x}) = 3.7 \times 10^{-3}$
↪  (true value: $V_{\text{true}}(\bar{x}) = 0$).
-
```

```
-\item \textbf{Curvature matching:} $\|\nabla^2 V_{\text{mid}}(\bar{x}) -
↪  P_{\text{loc}}\|_F = 4.2 \times 10^{-2}$, where $P_{\text{loc}}$ is the Riccati
↪  matrix at the equilibrium.
-\end{itemize}
-
-These metrics confirm that all constraints in the composite loss
↪  (\ref{eq:empirical-loss}) are satisfied to good accuracy, with small residuals
↪  reflecting the regularization trade-offs.
-
-
-\subsection{Example 2: SDE with Non-linear Drift and Local Potential Reconstruction}
-\label{sec:example2_kernels}
-
-In this example, we apply the hybrid kernel approach with the \emph{Warm Start}
-and \emph{Sparsity Refinement} strategies to a two-dimensional system with a
-non-linear drift, a case where the purely local quadratic approximation is
-accurate only within a single basin of attraction. We consider the double-well
-system
+
+\paragraph{Local constraints and interpretation.}
+The representation enforces $V(\bar x)=0$, $\nabla V(\bar x)=0$, and $D^2V(\bar x)=4I$
↪  algebraically. Accordingly, these three constraints do not rely on a
↪  finite-difference diagnostic or a penalty weight. The independent HJ checks
↪  evaluate the resulting field away from the collocation set. The example
↪  demonstrates recovery within a polynomial RKHS containing the target; general
↪  nonpolynomial systems require richer kernels and a separate assessment of
↪  approximation error.
+
+\FloatBarrier
+\subsection{Example 2: Nonlinear drift and local double-well
↪  reconstruction}\label{sec:example2_kernels}
+We consider
 \begin{equation}\label{sde2}
 \begin{aligned}
-    dx_1 &= (x_1 - x_1^3)\, dt + \sqrt{\varepsilon}\, dw_1, \\
-    dx_2 &= -x_2\, dt + \sqrt{\varepsilon}\, dw_2 ,
+ dx_1&=(x_1-x_1^3)\,dt+\sqrt\varepsilon\,dW_1,\\
+ dx_2&=-x_2\,dt+\sqrt\varepsilon\,dW_2.
 \end{aligned}
 \end{equation}
-\textcolor{red}{which has stable equilibria at $\bar{x}=(\pm 1, 0)$ and a saddle at
↪  $(0,0)$.} The associated quasi-potential is
-\[
-V_{\text{true}}(x)
-= \frac{1}{2} (x_1^2 - 1)^2 + x_2^2.
-\]
-\textcolor{red}{We use the same kernel dictionary as Example~1 (two Gaussian RBFs with
↪  $\sigma=0.5,1.5$ and five polynomial kernels of degree $r=2,\ldots,6$). In the
↪  results below, ``Riccati'' (abbreviated ``quad.'\ in tables) refers to the local
↪  quadratic approximation from the Riccati equation, and ``Hybrid'' (abbreviated
↪  ``Hyb.'\ or subscript $h$) refers to our proposed kernel surrogate.}
-
-\paragraph{Riccati Approximation and Warm Start.}
-We linearize the system at the stable equilibrium $\bar{x}=(1,0)$.
-The resulting local quadratic approximation has Hessian
-\[
-Z_{\text{ric}}
-=
-\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}.
-\]
```

```
-This Riccati matrix is used as a \emph{Warm Start} for the Hybrid Kernel
-optimization so that the surrogate potential automatically satisfies the exact
-local curvature constraint at $(1,0)$.
-
-\paragraph{Hybrid Kernel Results and Sparsity Analysis.}
-We employ a mixed dictionary of $L=7$ kernels (two RBFs and five polynomial
-kernels).
-The $\ell_1$ penalty does not prune any kernel: all seven kernels remain active,
-each with normalized weight $\approx 0.1429$. This reflects the nonlinearity of
-the true potential near $(1,0)$, which cannot be captured by a single quadratic
-term.
-
-Importantly, both the Riccati approximation and the Hybrid Kernel surrogate are
-constructed \emph{solely} from collocation points in the basin of attraction of
-$(1,0)$. Therefore both reconstructions are intrinsically \emph{local}:
-neither method receives any information about the second well at $(-1,0)$, and
-neither is expected to reconstruct it. The Hybrid Kernel method should
-therefore be interpreted as a refined local surrogate, not a global
-reconstruction, for this example.
-
-\begin{itemize}
-    \item Local Riccati approximation error:
-    \[
-      \mathrm{RMSE}_{\rm quad}=5.3443.
-    \]
-    \item Hybrid Kernel local approximation error:
-    \[
-      \mathrm{RMSE}_{h}=4.7418,
-      \qquad
-      \mathrm{Rel\ }L^2 \approx 0.79,
-    \]
-    computed on a uniform grid over $[-1.5,1.5]^2$.
-\end{itemize}
-
-\paragraph{Relative $L^2$ error and Hausdorff distances.}
-We report the relative $L^2$ error
-\[
-\mathrm{Rel}\,L^2
-=
-\frac{\|V_{\mathrm{approx}} - V_{\mathrm{true}}\|_{L^2(\Omega)}}
-     {\|V_{\mathrm{true}}\|_{L^2(\Omega)}},
-\]
-and the \emph{single-basin} Hausdorff distance between the level sets
-$\Gamma_c = \{x : V(x)=c\}$ restricted to the basin of $(1,0)$.
-Across all finite interior levels (up to the separatrix), we obtain
-\[
-d_H^{\mathrm{Ric}} \approx 7.412\times10^{-1},
-\qquad
-d_H^{\mathrm{Hyb}} \approx 7.391\times10^{-1},
-\]
-showing that both reconstructions reproduce the local geometry of the well with
-comparable fidelity. The improvement of the Hybrid Kernel method is modest in
-this gradient-Gaussian example, but becomes significant in the non-gradient,
-non-linear three-dimensional example in Section~\ref{sec:example3_kernels}.
-
-To quantify the performance of the two local reconstruction approaches, we report in
-Table~\ref{tab:example2_errors} three complementary error metrics-RMSE, relative
-$L^{2}$ error, and the single-basin Hausdorff distance between level sets. These
↪  values
```

```
-provide a direct numerical comparison between the purely quadratic Riccati
↪  approximation
-and the Hybrid Kernel surrogate, complementing the visual assessment in the preceding
-figures. As the table shows, both methods reproduce the local geometry of the
↪  potential
-well with comparable accuracy, with the Hybrid Kernel method offering a modest but
-consistent improvement across all reported metrics.
-
-\begin{table}[h!]
-    \centering
-    \caption{Error metrics for Example~2 (double-well system) comparing the
-    local Riccati approximation and the Hybrid Kernel surrogate, evaluated on a
-    uniform grid in $[-1.5,1.5]^2$ and restricted to the basin of the stable
-    equilibrium $(1,0)$.}
-    \label{tab:example2_errors}
-    \vspace{0.3em}
-    \begin{tabular}{lccc}
-        \toprule
-        Method & RMSE & Rel $L^2$ & Hausdorff $d_H$ \\
-        \midrule
-        Riccati (local quad.)  & $5.3443$ & $0.79$ & $7.412\times 10^{-1}$ \\
-        Hybrid Kernel          & $4.7418$ & $0.79$ & $7.391\times 10^{-1}$ \\
-        \bottomrule
-    \end{tabular}
-\end{table}
-
- The Hybrid Kernel
-method improves the local reconstruction of the basin around $(1,0)$, whereas
-the Riccati approximation remains purely quadratic.
-
-\paragraph{Constraint satisfaction diagnostics.}
-For the hybrid kernel approximation trained on $N = 200$ collocation points in the
↪  basin of $(1,0)$, we obtain:
-
-\begin{itemize}
-\item \textbf{Hamilton--Jacobi residual:} Mean $|\mathcal{L}_{\text{HJ}}| = 2.1 \times
↪  10^{-2}$ with maximum $9.5 \times 10^{-2}$ on the collocation set.
-
-\item \textbf{Boundary condition:} $V_h(1,0) = 2.8 \times 10^{-3}$ (true value:
↪  $V_{\text{true}}(1,0) = 0$).
-
-\item \textbf{Curvature matching:} $\|\nabla^2 V_h(1,0) - Z_{\text{ric}}\|_F = 3.6
↪  \times 10^{-2}$, where $Z_{\text{ric}} = \text{diag}(4, 2)$ is the Riccati Hessian.
-\end{itemize}
-
-As reported in Table~\ref{tab:example2_errors}, the local Riccati approximation and
↪  hybrid kernel surrogate achieve comparable geometric fidelity (Hausdorff distances
↪  $d_H^{\text{Ric}} \approx 7.41 \times 10^{-1}$ and $d_H^{\text{Hyb}} \approx 7.39
↪  \times 10^{-1}$), with the hybrid method offering a modest but consistent
↪  improvement in RMSE and relative $L^2$ error.
-
- \iffalse
- \subsection{Example 3: SDE with multiplicative noise and Quasipotential
↪  Reconstruction}
-\label{sec:example3_kernels}
-
-In this section, we present numerical experiments comparing (i) the classical
↪  quadratic approximation of the quasi-potential obtained from the linearization at
↪  the equilibrium (Riccati), and (ii) our proposed Hybrid Kernel Approximation
↪  utilizing a Warm Start and sparsity refinement.
```

```
-
-We consider the three-dimensional system with multiplicative noise defined by
-\begin{equation}\label{sde3}
+The stable equilibria are $(\pm1,0)$, separated by the saddle $(0,0)$. We reconstruct
↪  the potential relative to $\bar x=(1,0)$ in the right basin. The local Jacobian is
↪  $\operatorname{diag}(-2,-1)$, so the Riccati Hessian and quadratic are
+\[
+ P=\begin{pmatrix}4&0\\0&2\end{pmatrix},\qquad
+ V_Q(x)=2(x_1-1)^2+x_2^2.
+\]
+Within this basin the analytic potential is
+\begin{equation}\label{eq:dw-reference}
+ V_*(x)=\tfrac12(x_1^2-1)^2+x_2^2.
+\end{equation}
+Its continuation to $x_1<0$ is a useful polynomial reference but is not interpreted
↪  here as the global action value relative to the right-hand attractor.
+
+\paragraph{Training and reconstruction.}
+Training, validation, and independent random testing use $[0.05,1.5]\times[-1.5,1.5]$,
↪  entirely within the right basin. The continuation protocol and degree-selection
↪  rule are the same as in Example~1. Every training seed selects $r=4$. With
↪  $y_1=x_1-1$ and $y_2=x_2$, the principal fitted correction is
+\[
+ V_4(x)-V_Q(x)\approx 2y_1^3+\tfrac12y_1^4.
+\]
+Thus the selected polynomial space captures the nonlinear geometry with nine free
↪  correction coefficients. The data constrain the local well, while the reference
↪  formula permits a separate extrapolation test on a wider box.
+
+\input{results/example2_verified_table.tex}
+
+For seeds $42,43,44$, the independent-basin RMSEs are $5.22\times10^{-8}$,
↪  $4.86\times10^{-8}$, and $4.74\times10^{-8}$. The Riccati baseline has RMSE
↪  $0.4130$ on the same test set. The corresponding fitted relative errors are about
↪  $4\times10^{-8}$. On the wider $101^2$ grid over $[-1.5,1.5]^2$, the errors against
↪  the polynomial extension are $2.87\times10^{-7}$, $6.06\times10^{-7}$, and
↪  $6.17\times10^{-7}$. This separates interpolation within the training basin from
↪  extrapolation of the learned polynomial.
+
+\paragraph{Level-set geometry.}
+For each level $c$, we approximate its level set by the grid band
+\[
+ \Gamma_{c,h}(V)=\{x\in X_h:|V(x)-c|\leq0.02\},
+\]
+using a $101^2$ grid on the training box and levels $0.1,0.2,0.3,0.4$, below the
↪  saddle height $1/2$. The symmetric Hausdorff distance is computed between these
↪  finite point sets and then averaged over the four levels. The fitted and reference
↪  bands coincide on this grid, yielding zero discrete distance; this means agreement
↪  at the stated spatial and band resolution. It is not a claim of exact equality of
↪  the continuous level sets. The Riccati mean distance is approximately $0.1317$.
+
+\paragraph{Constraint and error assessment.}
+The equilibrium value, gradient, and Hessian are imposed exactly. For seed 42 the mean
↪  absolute HJ residual on the independent basin test is $6.15\times10^{-8}$ and its
↪  maximum is $6.29\times10^{-7}$. These diagnostics complement the potential error:
↪  local curvature describes the behavior near the stable point, the potential error
↪  measures the scalar reconstruction throughout the testing domain, and the level-set
↪  comparison measures its geometry. Their agreement here is facilitated by exact
↪  polynomial representability and noiseless access to the drift.
```

```
+
+\subsection{Example 3: Multiplicative noise and quadratic
↪  reconstruction}\label{sec:example3_kernels}
+Consider the three-dimensional SDE with $\rho=5$:
+\begin{equation}\label{eq:example3-sde}
 \begin{aligned}
-    dx_1 &= -x_1 (1+x_1^2)dt + \sqrt{\varepsilon (1+x_1^2)} dw_1, \\
-    dx_2 &= (-\frac{1}{2}x_2 - \rho x_3)dt + \sqrt{\varepsilon} dw_2, \\
-    dx_3 &= [-\frac{1}{2}x_3 + \rho x_2]dt + \sqrt{\varepsilon} dw_3
+ dx_1&=-x_1(1+x_1^2)\,dt+\sqrt{\varepsilon(1+x_1^2)}\,dW_1,\\
+ dx_2&=(-\tfrac12x_2-\rho x_3)\,dt+\sqrt\varepsilon\,dW_2,\\
+ dx_3&=(-\tfrac12x_3+\rho x_2)\,dt+\sqrt\varepsilon\,dW_3.
 \end{aligned}
 \end{equation}
-and its associated quasi-potential
-\begin{equation}\label{eq3:Q-def}
-V_{\text{true}}(x) = x_1^2 + \frac{1}{2} x_2^2 + \frac{1}{2} x_3^2,
-\end{equation}
-where $\rho=5$.
-
-Linearizing at $x_{\mathrm{eq}}=0$ gives the Jacobian
-\[
-  J = \nabla F(0)
-  = \begin{pmatrix}
-    -1 & 0 & 0\\
-    0  & -\tfrac12 & -\rho\\
-    0  &  \rho     & -\tfrac12
-  \end{pmatrix}.
-\]
-The standard quadratic quasi-potential $V_{\rm quad}$ is derived by solving the
↪  algebraic Riccati equation associated with this linearization. As a baseline, we
↪  compute the error of this quadratic approximation relative to the true potential
↪  over a uniform grid of $50^3$ points in $[-2,2]^3$. The baseline performance (after
↪  optimal scaling) is:
-\[
-  \mathrm{RMSE}_{\rm quad}=0.5064,\quad
-  \max|V_{\rm true}-\alpha_{\rm quad}V_{\rm quad}|=1.3328,\quad
-  \mathrm{Rel\ }L^2=0.1601,\quad
-  d_H=1.1547.
-\]
-
-\paragraph{Hybrid Kernel Approximation.}
-For our method, we use the sum of a dictionary of $L=7$ kernels: two Gaussian RBFs and
↪  five polynomial kernels of degree $r=2$ to $6$. We utilize a set of $N_{\rm
↪  col}=400$ random collocation points. To overcome the initialization challenges and
↪  poor convergence observed in initial runs, we implement a \textbf{Warm Start}
↪  strategy:
-\begin{enumerate}
-    \item The curvature penalty is enforced by setting the target Hessian
↪  $\nabla^2V(0)$ to the Riccati solution $Z$ (computed from the linearization).
-    \item The kernel coefficients $\alpha$ are initialized by performing a ridge
↪  regression to fit the sum of kernels to the local quadratic potential
↪  $V_{\text{quad}}(x) = \frac{1}{2}x^T Z x$ at the collocation points.
-\end{enumerate}
-We minimize the total objective (Equation \ref{eq:composite_loss}) with a sparsity
↪  parameter $\lambda_1 = 0.05$ and curvature penalty $\gamma=10$.
-
-\paragraph{Results and Sparsity Analysis.}
```

```
-The optimization and subsequent sparsity refinement (pruning kernels with weights
↪  below $2\%$) yielded a powerful model selection result: the weight of the flexible
↪  RBF kernels and the high-degree polynomial kernels ($r=3$ to $r=6$) were driven to
↪  zero. The final kernel weights were $100\%$ concentrated on the $\text{Poly}(r=2)$
↪  kernel. This confirms that the algorithm successfully "discovered" that the
↪  quasi-potential for this system is purely quadratic.
-
-The final evaluation of the refined Hybrid Kernel approximation $V_h$ on the same
↪  $50^3$ grid gives significantly improved metrics:
-\[
-  \mathrm{RMSE}_{h}=\mathbf{0.0010},\quad
-  \max|V_{\rm true}-V_h|=0.0026,\quad
-  \mathrm{Rel\ }L^2=0.0003,\quad
-  d_H=0.0026.
-\]
-These results demonstrate that the Hybrid Kernel method, when properly initialized, is
↪  superior to the standard Riccati approach and successfully performs automated model
↪  selection.
-
-Figure~\ref{fig:comparison} displays contour slices of the True Potential, the Riccati
↪  approximation, and our Hybrid Kernel approximation. The Hybrid Kernel result is
↪  visually indistinguishable from the ground truth.
-
-\begin{figure}[ht]
-  \centering
-  \includegraphics[width=\textwidth]{quasi_potential_comparison.png}
-  \caption{Comparison of the true quasi-potential (Left), the Riccati/Quadratic
↪  approximation (Middle), and the proposed Hybrid Kernel approximation (Right) on
↪  three orthogonal slices. The Hybrid Kernel method achieves an RMSE of $\approx
↪  0.001$, correctly recovering the quadratic nature of the potential.}
-  \label{fig:comparison}
+The equilibrium is zero, the diffusion is $a(x)=\operatorname{diag}(1+x_1^2,1,1)$, and
+\[
+ J=\begin{pmatrix}-1&0&0\\0&-1/2&-\rho\\0&\rho&-1/2\end{pmatrix}.
+\]
+Solving the Lyapunov equation at zero gives $P=\operatorname{diag}(2,1,1)$. The
↪  analytic potential is exactly the Riccati quadratic,
+\begin{equation}\label{eq:example3-reference}
+ V_*(x)=x_1^2+\tfrac12x_2^2+\tfrac12x_3^2.
+\end{equation}
+The rotational terms cancel in $b\cdot\nabla V_*$, while the state-dependent diffusion
↪  cancels the nonlinear first-coordinate drift in the HJ equation. This example tests
↪  diffusion handling, normalization, and numerical stability of the kernel
↪  representation. Since the Riccati warm start is already exact, it is a consistency
↪  benchmark rather than an improvement over that baseline.
+
+\paragraph{Anchored kernel fit.}
+We use the degree-two component of the dictionary,
+\[
+ V_h(x)=\sum_{j=1}^{N}\alpha_j\bigl[(1+x\cdot z_j)^2-1\bigr],
+\]
+with centers $z_j=x_j$ drawn uniformly from $[-2,2]^3$ by seed 42. The kernel weight
↪  is one, and the subtraction enforces $V_h(0)=0$. Training minimizes
+\begin{equation}\label{eq:quadratic-refit-loss}
+ \frac1N\sum_{p=1}^N H(x_p,\nabla V_h(x_p))^2
+ +10\|D^2V_h(0)-P\|_F^2+10^{-4}\|\alpha\|_2^2.
+\end{equation}
```

```
+The amplitude sparsity term is constant for this single active kernel and therefore
↪  does not affect the refit. The nine-dimensional feature parametrization in
↪  Section~\ref{sec:direct-implementation} eliminates coefficient redundancy while
↪  preserving the minimum-norm $\alpha$ and the scalar kernel function. Curvature is
↪  penalized rather than fixed in this experiment; the equilibrium gradient remains a
↪  fitted quantity. The warm start is computed from the local Lyapunov equation.
+
+\input{results/example3_verified_table.tex}
+
+\paragraph{Accuracy and constraints.}
+At $N=200$, the fixed $20^3=8000$ point grid on $[-2,2]^3$ gives RMSE
↪  $1.35\times10^{-9}$, with the maximum and relative errors in
↪  Table~\ref{tab:example3_comparison}. The Riccati baseline remains at floating-point
↪  precision. The value anchor is exact by construction. Hessian and gradient errors,
↪  HJ residuals, and independent random-test errors are computed directly from the
↪  fitted coefficients. The supplied implementation verifies the equality of
↪  feature-space predictions and the original anchored kernel sum at independent
↪  points, as well as the analytic residual Jacobian by directional finite
↪  differences.
+
+\paragraph{Sample-count study.}\label{sec:empirical-convergence}
+We repeat the fit for $N=50,100,200,400,800$, using the same seeded sampling rule and
↪  fixed evaluation grid. The kernel-refit RMSEs are respectively $1.05\times10^{-7}$,
↪  $5.95\times10^{-9}$, $1.35\times10^{-9}$, $1.13\times10^{-9}$, and
↪  $3.88\times10^{-10}$. Every solve satisfies its gradient stopping criterion; the
↪  maximum final first-order optimality measure is below $10^{-8}$.
↪  Figure~\ref{fig:convergence_study} also gives errors on an independent set of
↪  $10000$ uniform points and the HJ residuals on both evaluation sets.
+
+The number of identifiable polynomial coefficients is nine for all these values of
↪  $N$. Increasing $N$ changes the collocation average and the minimum-norm
↪  coefficient representation used in the ridge term. Consequently, the displayed
↪  trend describes numerical and regularization sensitivity within a fixed
↪  finite-dimensional model; no power-law approximation rate is fitted to it. The
↪  nearly exact quadratic warm start is central to this behavior.
+
+\begin{figure}[htbp]\centering
+\includegraphics[width=\linewidth]{figures/native/convergence_study_example3.pdf}
+\caption{Anchored degree-two kernel fits for the multiplicative-noise system. Left:
↪  potential RMSE; right: mean absolute HJ residual. The solid curves use the fixed
↪  $20^3$ grid and the dashed curves use $10000$ independent random points. Seed 42
↪  and the same objective are used for every $N$.}\label{fig:convergence_study}
 \end{figure}
-\fi
-
-\subsection{Example 3: SDE with multiplicative noise and quasi-potential
↪  reconstruction}
-\label{sec:example3_kernels}
-
-In this example we consider a three-dimensional system with multiplicative noise,
-\begin{equation}\label{eq:example3_system}
-\begin{aligned}
-    dx_1 &= -x_1(1+x_1^2)\,dt + \sqrt{\varepsilon(1+x_1^2)}\,dw_1, \\
-    dx_2 &= \left(-\tfrac12 x_2 - \rho x_3\right)dt + \sqrt{\varepsilon}\,dw_2,\\
-    dx_3 &= \left(-\tfrac12 x_3 + \rho x_2\right)dt + \sqrt{\varepsilon}\,dw_3 ,
-\end{aligned}
-\end{equation}
```

```
-\textcolor{red}{with $\rho = 5$ and equilibrium $\bar{x}=0$. We again use the same
↪  kernel dictionary as Examples~1 and~2 (two Gaussian RBFs and five polynomial
↪  kernels of degree $r=2,\ldots,6$).}
-The associated quasi-potential is known in closed form:
-\begin{equation}\label{eq:example3_true_potential}
-V_{\mathrm{true}}(x)
-= x_1^2 + \tfrac12 x_2^2 + \tfrac12 x_3^2 .
-\end{equation}
-
-\paragraph{Quadratic baseline (Riccati).}
-Linearizing the drift at $\bar{x}=0$ yields the Jacobian
-\[
-J
-=
-\begin{pmatrix}
--1 & 0 & 0 \\
-0 & -\tfrac12 & -\rho\\
-0 & \rho & -\tfrac12
-\end{pmatrix},
-\qquad
-A(0)=\mathrm{diag}(1,1,1).
-\]
-Solving the Lyapunov equation $JX+XJ^\top = -A(0)$ gives the covariance matrix
-$X$, and the Riccati Hessian is $Z = X^{-1}$.
-Thus the classical local approximation is
-\[
-V_{\mathrm{quad}}(x)=\tfrac12\,x^\top Z\,x.
-\]
-Because $V_{\mathrm{true}}$ is exactly quadratic, the Riccati approximation
-matches it up to machine precision.
-Evaluating on a $20^3$ grid in $[-2,2]^3$ gives:
-\[
-\mathrm{RMSE}_{\mathrm{quad}}
-= 3.4\times 10^{-16},\quad
-\mathrm{Rel}\,L^2 = 1.0\times 10^{-16},\quad
-d_H = 0.
-\]
-
-% ADD after the paragraph "Riccati Approximation and Warm Start..."
-% and BEFORE "We minimize the empirical loss..."
-\iffalse
-\textbf{Initialization procedure.} To ensure stable optimization and avoid convergence
↪  to poor local minima, we initialize the kernel coefficients $\alpha^{(i)}$ using
↪  ridge regression to approximate the Riccati quadratic $\frac{1}{2}x^\top Z x$ at
↪  the collocation points. Specifically, for each eigenfunction $\phi_i$, we solve:
-\[
-\min_{\alpha^{(i)}} \sum_{p=1}^{N_{\text{col}}} \left| \sum_{j=1}^{N_{\text{col}}}
↪  \alpha^{(i)}_j k_\beta(x_p, x_j) + (x_p - 0) \cdot v_i - \sqrt{x_p^\top Z x_p}
↪  \right|^2 + \eta \|\alpha^{(i)}\|_2^2,
-\]
-where $v_i$ are the eigenvectors of the Jacobian $J = \nabla b(0)$ and $Z$ is the
↪  Riccati Hessian. This warm-start strategy provides coefficients that already
↪  approximately satisfy the curvature constraint $\nabla^2 V(0) \approx Z$ before the
↪  full composite loss optimization begins, preventing the pathological convergence
↪  behavior observed with random initialization. After warm-start initialization, we
↪  proceed with the full optimization of the composite loss~\eqref{eq:composite_loss}.
-\fi
-
-\paragraph{Hybrid Kernel approximation (Warm Start + Sparsity).}
-We now apply the Hybrid Kernel method using a dictionary of $L=7$ kernels
```

```
-(two RBFs and five polynomial kernels of degree $r=2,\dots,6$), with
-$N_{\mathrm{col}}=200$ random collocation points.
-A \emph{Warm Start} is used to ensure stable optimization:
-\begin{enumerate}
-    \item The curvature penalty enforces the Hessian constraint
-    $\nabla^2 V(0)=Z$ from the Riccati solution.
-    \item The kernel coefficients $\alpha$ are initialized by ridge regression
-    to match the quadratic function $\tfrac12 x^\top Z x$ at the collocation points.
-\end{enumerate}
-We minimize the empirical loss~\eqref{eq:composite_loss}
-with curvature weight $\gamma=10$ and sparsity parameter $\lambda_1=0.05$.
-After optimization we prune kernels with weight below $2\%$.
-
-\paragraph{Sparsity and automatic model selection.}
-A remarkable outcome of the sparsity-promoting optimization is that \textit{all}
-flexible kernels (RBFs and polynomial degrees $r = 3, 4, 5, 6$) are eliminated
-during optimization. After convergence and pruning kernels with $\beta_j < 0.02$,
-we obtain:
-\begin{equation}\label{eq:example3_sparsity}
-\beta_j^2 =
-\begin{cases}
-\approx 1.00, & \text{Polynomial degree } r=2, \\
-0, & \text{otherwise.}
-\end{cases}
-\end{equation}
-Thus, the Hybrid Kernel method \textit{automatically discovers} the correct
-quadratic structure of the quasi-potential through the $\ell_1$ penalty, without
-prior knowledge that the true solution~\eqref{eq:example3_true_potential} is
-quadratic.
-
-\paragraph{Accuracy of the final reconstruction.}
-After sparsification, the hybrid approximation $V_h$ remains extremely close to
-the ground truth, though not identical due to the trade-offs between PDE residual
-minimization, curvature matching, and sparsity in the composite
-loss~\eqref{eq:composite_loss}. Evaluated on a $20^3 = 8000$ point test grid in
-$[-2, 2]^3$, we obtain:
-\begin{equation}\label{eq:example3_errors}
-\text{RMSE}_h \approx 1.0 \times 10^{-3}, \quad
-\max|V_{\text{true}} - V_h| \approx 3.5 \times 10^{-3}, \quad
-\text{Rel}\,L^2 \approx 3.2 \times 10^{-4}, \quad
-d_H \approx 0.
-\end{equation}
-
-Table~\ref{tab:example3_comparison} summarizes the comparison. The Riccati
-quadratic is exact up to machine precision, as expected. The Hybrid Kernel
-method successfully identifies the underlying structure and achieves high accuracy
-with automatic kernel selection.
-
-\begin{table}[htbp]
-\centering
-\caption{Comparison of reconstruction accuracy for Example~3
-(Equation~\eqref{eq:example3_system}). The Riccati quadratic is exact; the
-Hybrid Kernel method automatically selects the quadratic kernel and yields a
-near-perfect approximation.}
-\label{tab:example3_comparison}
-\vspace{0.4em}
-\begin{tabular}{lcccc}
-\toprule
-Method & RMSE & Max AE & Rel $L^2$ & Avg.\ Hausdorff $d_H$ \\
-\midrule
```

```
-Riccati (quadratic) & $3.4\times 10^{-16}$ & $1.8\times 10^{-15}$ & $1.0\times
↪  10^{-16}$ & $0$ \\
-Hybrid Kernel (sparse) & $1.0\times10^{-3}$ & $3.5\times10^{-3}$ & $3.2\times10^{-4}$
↪  & $\approx 0$ \\
-\bottomrule
-\end{tabular}
-\end{table}
-
-\paragraph{Constraint satisfaction diagnostics.}
-To verify constraint enforcement, we report the following diagnostics for the
-optimized $V_h$:
-
-\begin{itemize}
-\item \textbf{Hamilton--Jacobi residual:}
-\begin{equation}\label{eq:HJ_residual}
-\mathcal{L}_{\text{HJ}}(x) = b(x)^\top \nabla V_h(x) + \tfrac{1}{2} \nabla V_h(x)^\top
↪  a(x) \nabla V_h(x).
-\end{equation}
-On the collocation grid: Mean $|\mathcal{L}_{\text{HJ}}| = 2.3 \times 10^{-3}$,
-Max $|\mathcal{L}_{\text{HJ}}| = 8.7 \times 10^{-3}$.
-
-\item \textbf{Boundary condition:} $V_h(\mathbf{0}) = 1.2 \times 10^{-4}$
-(true value: $V(\mathbf{0}) = 0$).
-
-\item \textbf{Curvature matching:} $\|\nabla^2 V_h(\mathbf{0}) - Z\|_F = 5.8 \times
↪  10^{-3}$,
-where $Z$ is the Riccati matrix. Before optimization this error was $0.82$,
-demonstrating effective enforcement of the curvature constraint.
-\end{itemize}
-
-These diagnostics confirm that all constraints are satisfied to good accuracy,
-with small but non-zero residuals reflecting the regularization trade-offs
-in~\eqref{eq:composite_loss}.
-
-
-\paragraph{Convergence as a function of $N$.}
-To verify the theoretical convergence guarantee established in
-Theorem~\ref{thm:consistency-curvature}, we conducted a systematic study of
↪  approximation
-accuracy as a function of the number of collocation points $N$.
-\textcolor{red}{Figure~\ref{fig:convergence_study} displays the RMSE between the
↪  Hybrid Kernel
-approximation and the analytic quasi-potential for $N \in \{50, 100, 200, 400, 800,
↪  1600, 3200, 6400\}$,
-evaluated on the same $20^3 = 8000$ point test grid. Results are reported as the mean
-over 10 independent random seeds, with error bars indicating one standard deviation.}
-
-On the log-log scale (Figure~\ref{fig:convergence_study}, left panel), the RMSE
-exhibits clear power-law decay:
-\begin{equation}\label{eq:convergence_rate}
-\text{RMSE} \sim C \cdot N^{-\alpha}, \quad \alpha \approx 0.73,
-\end{equation}
-corresponding to approximately 65\% error reduction when doubling the number of
-collocation points. The fitted curve (dashed line) closely matches the observed
-data across the entire range, confirming consistent convergence without stagnation.
-The semi-log plot (right panel) provides an alternative visualization showing
-monotonic decay toward the Riccati baseline.
-
-The Riccati quadratic approximation (orange squares) achieves machine precision
-(RMSE $\approx 3 \times 10^{-16}$) independent of $N$, as expected, since the
```

```
-true quasi-potential~\eqref{eq:example3_true_potential} is exactly quadratic for
-this system. The Hybrid Kernel method successfully discovers this structure
-through automatic kernel selection (see Equation~\eqref{eq:example3_sparsity}),
-pruning all non-quadratic terms and converging toward the same optimum.
-
-These results directly support the finite-sample error bounds derived in
-Theorem~\ref{thm:consistency-curvature} and demonstrate that increased collocation
↪  density
-leads to improved approximation quality in practice. The observed rate
-$\alpha \approx 0.73$ is consistent with convergence rates typically achieved by
-kernel-based collocation methods for smooth solutions on bounded domains.
+
+\paragraph{Potential slices and spatial errors.}
+Figures~\ref{fig:comparison_example3} and~\ref{fig:residuals_example3} evaluate the
↪  three coordinate planes through the equilibrium. Each plane uses an $81\times81$
↪  grid containing zero. The displayed reference, Riccati, and fitted contours
↪  coincide visually. Absolute-error panels use separate labeled scales to resolve the
↪  kernel's small regularization bias and the Riccati baseline's rounding error. No
↪  error offset or fitted amplitude is introduced.
+
+\begin{figure}[htbp]\centering
+\includegraphics[width=0.95\linewidth,height=0.73\textheight,keepaspectratio]{figures
↪  /native/quasi_potential_slices_sparse.pdf}
+\caption{Coordinate-plane slices of the analytic potential, Riccati baseline, and
↪  anchored degree-two kernel fit at $N=200$. Every fixed coordinate is exactly zero.
↪  A common potential range is used across all panels.}\label{fig:comparison_example3}
+\end{figure}
+\begin{figure}[htbp]\centering
+\includegraphics[width=0.95\linewidth]{figures/native/residuals_example3.pdf}
+\caption{Absolute errors on the zero-coordinate slices in
↪  Figure~\ref{fig:comparison_example3}. The top row is the Riccati baseline and the
↪  bottom row the anchored kernel fit. Each row uses its own labeled error
↪  scale.}\label{fig:residuals_example3}
+\end{figure}
+
+\paragraph{Reproducibility.}
+The computational package stores the training points, fitted coefficients, validation
↪  scores for every candidate, independent test points, solver termination data, and
↪  numerical derivatives. The programs \path{code/run_polynomial_experiments.py} and
↪  \path{code/refit_selected_kernel.py} generate the two-dimensional and
↪  three-dimensional fits, respectively. The program
↪  \path{code/plot_verified_experiments.py} generates all seven vector-PDF figures and
↪  both example tables directly from those results. The reference formulas are
↪  evaluated after model fitting and selection. Each figure is therefore reproducible
↪  without raster inputs or manual curve tracing.
+
+\FloatBarrier
+\section{Conclusion}
+
+We have presented a kernel framework for quasi-potential computation with direct
↪  scalar and Koopman-coordinate representations. Both use Hamilton--Jacobi structure
↪  and local Riccati curvature. The spectral form supplies dynamical coordinates and a
↪  nonnegative quadratic surrogate, whereas direct scalar fitting avoids the
↪  additional transport restriction. Nonnegative amplitude regularization and discrete
↪  dictionary selection offer distinct controls on model complexity.
+
```

```
+Theorem~\ref{thm:consistency-curvature} gives sufficient conditions for convergence to
↪  the selected viscosity solution. The numerical experiments recover the
↪  two-dimensional quartic benchmark potentials using degree-four kernel spaces and
↪  the multiplicative-noise quadratic using an anchored degree-two expansion.
↪  Independent validation and test points assess residuals and potential errors.
↪  Finite-feature compression stabilizes the quadratic kernel refit without changing
↪  its function class. The controlled Gaussian-kernel experiments include a structured
↪  eight-dimensional system and favor direct fitting over the tested restricted
↪  spectral candidate. These findings identify when the global representational
↪  restriction, rather than the local curvature information, controls approximation
↪  accuracy.
+
+\paragraph{Limitations and future work.}
+The main limitation of the current work is the absence of a head-to-head comparison
↪  with established methods such as ordered-upwind methods~\cite{cameron2012finding,
↪  dahiya2018ordered}, string methods~\cite{heymann2008geometric,
↪  grafke2019numerical}, and geometric minimum action
↪  approaches~\cite{zhou2008adaptive}. Such a comparison, which would require careful
↪  tuning of each method on common test problems, is left as an important direction
↪  for future work.
+Dense assembly and factorization have different costs, detailed in
↪  Section~\ref{sec:complexity-full}; the present evidence does not establish a
↪  universal practical dimension cutoff; sparse collocation, reduced-order modeling,
↪  and hierarchical kernel architectures are promising mitigation strategies.
↪  Extensions to non-Gaussian or Lévy noise, globally consistent multi-basin
↪  reconstruction, and the fully data-driven setting (where drift and diffusion must
↪  themselves be estimated) also remain open.
+
+\section*{Acknowledgments}
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↪  (Grant Nos. 12302035, 12572038, and 12372030) and Fundamental Research Funds for
↪  the Central Universities (Grant No. 30925010411).
+
+\clearpage
+\appendix
+\section{Controlled comparisons and cost measurements}\label{app:additional}
+\subsection{Models and fitting protocol}\label{app:protocol}
+This appendix compares direct and quadratic spectral representations and measures
↪  their computational costs. Both models use a single Gaussian kernel with bandwidth
↪  selection, coefficient ridge regularization, and exact local constraints. They
↪  complement the finite polynomial-kernel implementations in Section~\ref{examples}.
↪  The restricted quadratic Koopman candidate studied here is one specific spectral
↪  model; its performance does not characterize every hybrid kernel construction.
+
+Let $y=x-\bar x$ and $P$ be the Riccati Hessian. For kernel centers $z_j$, subtract
↪  Taylor jets at the equilibrium:
+\begin{align}
+ \psi_j^{(1)}(x)&=k(x,z_j)-k(\bar x,z_j)-D_1k(\bar x,z_j)y,\\
+ \psi_j^{(2)}(x)&=\psi_j^{(1)}(x)-\tfrac12y^\top D_1^2k(\bar x,z_j)y.
+\end{align}
+These functions belong to an augmented kernel-plus-polynomial feature space. The
↪  direct scalar and restricted spectral candidates are
+\begin{align}\label{eq:appendix-models}
+ V_D(x)&=\tfrac12y^\top Py+\sum_{j=1}^M c_j\psi_j^{(2)}(x),\\
+ \Phi(x)&=y+\sum_{j=1}^M c_j^K\psi_j^{(1)}(x),\qquad
+ V_K(x)=\tfrac12\Phi(x)^\top P\Phi(x).
+\end{align}
```

```
+Both satisfy $V(\bar x)=0$, $\nabla V(\bar x)=0$, and $D^2V(\bar x)=P$ exactly. $V_K$
↪  is nonnegative; global nonnegativity of $V_D$ is not automatic. The direct
↪  coefficients minimize mean squared HJ residual plus $10^{-8}\|c\|_2^2$. The
↪  spectral coefficients minimize the mean squared real vector transport residual
↪  $\|D\Phi\,b-F\Phi\|_2^2$ plus $10^{-8}\|C_K\|_F^2$. The latter is linear least
↪  squares in $Mn$ coefficients. The Euclidean coefficient penalty differs from the
↪  RKHS norm defined by the kernel Gram matrix.
+
+For each system and each seed in $\{230923,230924,230925\}$, the methods share $N=256$
↪  independent uniform training points, $512$ validation points, $4096$ test points,
↪  and the first $M=32$ training points as centers. The kernel is Gaussian with
↪  $\ell/\sqrt n\in\{0.5,1,2\}$. Each method chooses its bandwidth by the smallest
↪  validation HJ residual; reference potential values never enter fitting or
↪  selection. Direct nonlinear least squares starts at the Riccati quadratic, uses an
↪  analytic Jacobian, tolerances $10^{-10}$, and a $200$-evaluation limit. The
↪  spectral fit uses the real coupled linear system. All candidate solves terminated
↪  successfully; the largest observed nonlinear evaluation count was $54$. The
↪  comparison uses common data and constraints, but not equal parameter counts or
↪  identical fitting objectives.
+
+\subsection{Domains, higher-dimensional example, and results}\label{app:accuracy}
+The first three systems use the drift and diffusion coefficients in
↪  Section~\ref{examples}, with local testing boxes $[-1.5,-0.5]\times[-0.6,0.6]$ for
↪  Example~1, $[0.5,1.5]\times[-0.6,0.6]$ for Example~2, and $[-1,1]^3$ for Example~3.
↪  These local domains and Gaussian fitting protocols differ from the
↪  polynomial-kernel experiments in Section~\ref{examples}. Example~3's Riccati
↪  quadratic is exact and is the direct model's starting point; its direct result is
↪  therefore a consistency check, not a learned improvement over Riccati.
+
+The eight-dimensional system is defined by
+\begin{equation}\label{eq:appendix-eight}
+ V_*(x)=\sum_{i=1}^8(x_i^2+0.1x_i^4),\quad
+ b=-\tfrac12\nabla V_*+R\nabla V_*,\quad a=I,
+\end{equation}
+with $R_{i,i+1}=1/4$, $R_{i+1,i}=-1/4$ and all other entries zero, on $[-1,1]^8$. Its
↪  equilibrium is zero, $F=-I+2R$ is Hurwitz, and $P=2I$. Skew symmetry gives
↪  $H(x,\nabla V_*)=0$. Completion of squares in the action gives $S_T\geq
↪  V_*(x)-V_*(0)$; the uphill flow $\dot x=\nabla V_*/2+R\nabla V_*$ attains equality
↪  in the infinite-time limit. Its reverse decreases the coercive $V_*$ strictly away
↪  from zero and converges to zero, which identifies the action value through
↪  truncation near the equilibrium. The drift is nonlinear and non-gradient, but the
↪  potential is separable, so this is a structured stress test rather than general
↪  high-dimensional validation.
+
+\input{results/accuracy_table.tex}
+
+Table~\ref{tab:accuracy} reports means and sample standard deviations across three
↪  seeds. Relative error is $\|u-V\|_{\ell^2(X_t)}/\|V\|_{\ell^2(X_t)}$; HJ RMS is the
↪  root mean squared residual on the independent test points. The direct model is more
↪  accurate than the restricted spectral candidate in all four tests. The double-well
↪  direct fit has appreciable seed variability, as shown by the tabulated standard
↪  deviation. The eight-dimensional mean relative errors are approximately $5.95\%$
↪  for Riccati, $1.38\%$ for direct fitting, and $16.31\%$ for the spectral candidate.
↪  These comparisons do not establish universal model rankings, a viscosity
↪  convergence certificate, or certified escape-barrier accuracy. They show why
↪  Koopman features should be treated as an optional restriction, rather than presumed
↪  to improve a direct fit.

 \begin{figure}[htbp]
```

```
 \centering
-\includegraphics[width=\textwidth]{convergence_study_example3.png}
-\caption{Convergence of the Hybrid Kernel method as a function of the number
-of collocation points~$N$ for Example~3 (Equation~\eqref{eq:example3_system}).
↪  \textcolor{red}{
-\textbf{Left:} RMSE vs.~$N$ on log-log scale for the Hybrid Kernel method
-(blue circles, mean $\pm$ 1 s.d.\ over 10 seeds), showing power-law decay
-$\text{RMSE} \sim N^{-0.73}$ (dashed fitted line).
-\textbf{Right:} Same data on semi-log scale. The Riccati quadratic baseline (orange
↪  squares)
-is shown separately and achieves machine precision since the true potential is exactly
↪  quadratic;
-its convergence curve is therefore plotted on a separate panel to avoid suppression of
↪  scale.}}
-\label{fig:convergence_study}
+\includegraphics[width=\textwidth]{figures/ablation.pdf}
+\caption{Controlled model comparison: mean relative potential error and
↪  independent-test HJ RMS over three seeds. MS, DW, Mult., and Rot. denote the
↪  Maier--Stein, double-well, multiplicative-noise, and eight-dimensional rotational
↪  systems. Values below $10^{-15}$ are drawn at $10^{-15}$ for visibility. All fits
↪  use the single-kernel protocol of
↪  Appendix~\ref{app:protocol}.}\label{fig:additional-ablation}
 \end{figure}

-The observed convergence rate $\alpha \approx 0.73$ represents the \emph{optimal
↪  scaling} of the approximation error with respect to the number of collocation
↪  points, meaning that this rate is consistent with the best achievable rates for
↪  kernel-based methods under our smoothness assumptions. This rate indicates that the
↪  error decreases by approximately 65\% when the number of collocation points is
↪  doubled, providing a predictable and quantifiable measure of the computational
↪  effort required to achieve a desired accuracy level. \textcolor{red}{The dominant
↪  computational cost is that of constructing and solving the kernel Gram system,
↪  which scales as $O(N^2 n)$ for $N$ collocation points in $n$ dimensions; this
↪  limits practical applicability to moderate dimensions ($n \lesssim 8$) with current
↪  implementations.}
-
-\paragraph{Visualization.}
-Figure~\ref{fig:comparison_example3} shows orthogonal contour slices of
-$V_{\mathrm{true}}$, the Riccati approximation, and the Hybrid Kernel
-reconstruction.
-The Riccati and true potentials coincide exactly; the Hybrid Kernel solution is
-visually nearly indistinguishable, confirming that the method successfully
-identifies the underlying quadratic structure. Note that the Riccati approximation
-achieves near-perfect accuracy ($\text{RMSE} \approx 10^{-16}$) because the true
-quasi-potential is exactly quadratic for this system-a consequence of the linear
-drift and constant diffusion in the $x_2$-$x_3$ subsystem and the special structure
-of the multiplicative noise in the $x_1$ component.
-
-\begin{figure}[htbp]
-  \centering
-  \includegraphics[width=\textwidth]{quasi_potential_slices_sparse.png}
-  \caption{Contour slices of the true quasi-potential (Left), the Riccati
-  quadratic approximation (Middle), and the Hybrid Kernel approximation (Right)
-  for Example~3. After sparsification, the Hybrid Kernel isolates the quadratic
-  kernel and reconstructs a near-perfect potential. \textcolor{red}{The corresponding
↪  spatial
-  residuals $|V_h - V_{\mathrm{true}}|$ are shown in
↪  Figure~\ref{fig:residuals_example3}.}}
-  \label{fig:comparison_example3}
```

```
-\end{figure}
-
-% Figure showing spatial residuals for Example 3
-\begin{figure}[htbp]
-  \centering
-  \includegraphics[width=\textwidth]{residuals_example3.png}
-  \caption{\textcolor{red}{Spatial residuals $|V_h - V_{\mathrm{true}}|$ for Example~3
↪  on three orthogonal slices ($x_3=0$, $x_1=0$, $x_2=0$). Top row: Riccati
↪  (quadratic) approximation residuals, which achieve RMSE $\approx 10^{-3}$ since the
↪  true quasi-potential is exactly quadratic and the Riccati matrix captures this
↪  structure precisely. Bottom row: Hybrid Kernel approximation residuals, matching
↪  the Riccati accuracy (RMSE $\approx 10^{-3}$) after sparsification. Starting from a
↪  dictionary of 19 features (monomials up to degree~3), the $\ell_1$ penalty
↪  automatically zeroes out all but three terms---precisely $x_1^2$, $x_2^2$, and
↪  $x_3^2$---recovering the true quadratic structure without prior knowledge of its
↪  form. Separate colorscales are used for each row; both are on the order of
↪  $10^{-3}$, confirming that errors are negligible throughout the domain.}}
-  \label{fig:residuals_example3}
-\end{figure}
-
-\section{Conclusion}
-
-\textcolor{red}{We have presented a kernel-based framework for computing
↪  quasi-potentials in stochastic dynamical systems. The method parameterizes Koopman
↪  eigenfunctions via RKHS ansätze and minimizes a composite
↪  loss~\eqref{eq:composite_loss} that simultaneously enforces the Hamilton--Jacobi
↪  equation, matches the local Hessian to the Riccati solution, and promotes kernel
↪  sparsity through $\ell_1$ regularization. The result is an analytical surrogate
↪  that is cheap to evaluate and automatically selects the correct functional form
↪  through sparsification.}
-
-\textcolor{red}{Theorem~\ref{thm:consistency-curvature} establishes consistency of the
↪  learned surrogate with the true quasi-potential as the collocation count $N$ grows,
↪  with convergence guaranteed in function values, gradients, and equilibrium
↪  curvature. Three benchmark examples validate the approach: Example~1
↪  (two-dimensional gradient system) demonstrates kernel sparsification; Example~2
↪  (double-well with non-trivial topology) shows that the hybrid surrogate improves
↪  upon the Riccati baseline; and Example~3 (three-dimensional multiplicative noise)
↪  confirms automatic recovery of the exact quadratic structure (RMSE $\approx
↪  10^{-3}$, matching the Riccati baseline) and an empirical convergence rate
↪  $\text{RMSE}\sim N^{-0.73}$ consistent with kernel collocation theory.}
-
-\paragraph{Limitations and future work.}
-\textcolor{red}{The main limitation of the current work is the absence of a
↪  head-to-head comparison with established methods such as ordered-upwind
↪  methods~\cite{cameron2012finding, dahiya2018ordered}, string
↪  methods~\cite{heymann2008geometric, grafke2019numerical}, and geometric minimum
↪  action approaches~\cite{zhou2008adaptive}. Such a comparison, which would require
↪  careful tuning of each method on common test problems, is left as an important
↪  direction for future work.
-Computational cost scales as $O(N^2 n)$, limiting applicability beyond $n\approx 5$--8
↪  dimensions with current implementations; sparse collocation, reduced-order
↪  modeling, and hierarchical kernel architectures are promising mitigation
↪  strategies. Extensions to non-Gaussian or Lévy noise, globally consistent
↪  multi-basin reconstruction, and the fully data-driven setting (where drift and
↪  diffusion must themselves be estimated) also remain open.}
-
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↪  (Grant Nos. 12302035, 12572038, and 12372030) and Fundamental Research Funds for
↪  the Central Universities (Grant No. 30925010411).
+\subsection{Runtime, memory accounting, and reproducibility}\label{app:runtime}
+\input{results/timing_table.tex}
+\input{results/dense_table.tex}
+
+Table~\ref{tab:timing} measures fitting/selection over all three bandwidths and
↪  value/gradient prediction at $4096$ test points. Initial model/Lyapunov setup,
↪  imports, plots, and reference-error calculation are excluded. Table~\ref{tab:dense}
↪  separately measures Gaussian Gram assembly and Cholesky factorization, using
↪  $10^{-6}I$ regularization and medians of three repeats after one warm-up. It is not
↪  a timing of the complete PDE solver. Gram storage is $8N^2$ bytes; peak process
↪  memory was not measured and would also include factors, distances, derivatives, and
↪  optimization work arrays. A single $6400\times6400$ float64 matrix already occupies
↪  $312.5$ MiB.
+
+The recorded environment is CPU float64 on macOS 15.8 ARM64, Python 3.12.14, NumPy
↪  2.5.3, SciPy 1.18.1, and Matplotlib 3.11.2. Single-thread limits were requested
↪  through environment variables and \texttt{threadpoolctl}; no inspectable pools were
↪  reported, so the vendor thread count was not independently confirmed. Short timings
↪  depend on system load and caching. Four matrix sizes do not establish a measured
↪  asymptotic complexity exponent.
+
+The computational archive contains \path{code/run_experiments.py}, which defines the
↪  systems, derivatives, seeds, data splits, and solvers for the controlled
↪  comparisons. It checks gradients and equilibrium Hessians by finite differences and
↪  verifies the analytic HJ/Riccati identities and coupled transport assembly before
↪  fitting. The file \path{results/experiments.json} contains all 36 method/seed
↪  outcomes, candidate diagnostics, and timing/environment metadata, while
↪  \path{results/fitted_models.npz} stores coefficients, centers, and plotted grids.
↪  The scripts \path{code/make_tables.py} and \path{code/generate_figures.py} produce
↪  the tables and vector-PDF comparison plots from saved results. These materials
↪  support reproduction of the Gaussian-kernel comparisons. The main-text polynomial
↪  experiments are generated by \path{code/run_polynomial_experiments.py} and
↪  \path{code/refit_selected_kernel.py}; their saved inputs and outputs are in
↪  \path{results/accuracy/}. The program \path{code/plot_verified_experiments.py}
↪  generates the seven main-text figures and two numerical tables.
+
+\FloatBarrier

 \bibliographystyle{abbrv}
-\bibliography{references,references_quasi_potential}
+\bibliography{references_full_revision}

 \end{document}
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